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TOPICAL REVIEW

A Review on Symbolic Path Integral Calculus and its Applications to Quantum Theories

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Abstract

Path integrals are a powerful and general formalism for the quantisation of classical theories. Even in cases where precise quantisation procedures are still unknown, the path integrals of the theory can still be *symbolically* written and used to establish formal proofs or heuristic arguments, sometimes for very general classes of quantum theories. Currently available textbooks, lectures and reviews concentrate on useful but specific technical aspects of path integrals, such as time-slicing and perturbative expansions, and thus usually overlook the ability of path integrals to be used for symbolic calculus. The present article offers a self-contained review of path integrals as formal objects, providing concepts (functional integration, formal propagators, time-ordering,...), formulae (to obtain wavefunctions, matrix elements,...), rules (functional integration by parts, long-time limit, changes of variables,...) and methods (lattice-slicing, semi-classical limit and perturbative expansions,...) that can be used when manipulating symbolic path integrals. We explain how many of these methods can be understood as special instances of the general framework of effective theories, which is naturally described by path integration. The symbols, rules and methods are consistently given physical interpretation so as to provide intuition on how path integrals formally overcome the problem of quantising classical theories. When they are known, the mathematical justification and limitations of each of the rules and methods are discussed. A number of important results are symbolically derived along the way in a self-contained manner (such as the Schwinger-Dyson equations, Heisenberg commutation relations, Ward-Takahashi identities, the Hellmann-Feynman theorem,...), and the physical interpretation of the symbolic derivation is emphasised.

Contents

Introduction **2**

1 Symbolic path integrals in quantum physics **3**
1.1 Introduction to symbolic path integrals . . . 3
1.2 Transition amplitudes and boundary conditions . . . 4
1.3 Notions of functional calculus . . . 6
1.4 Observable functionals and Schwinger-Dyson equations . . . 7
1.5 Various examples of target spaces . . . 10
1.6 Wick rotation and statistical physics . . . 11

2 Mathematical approaches to path integrals **12**
2.1 The Wiener measure and the Feynman-Kac formula . . . 13
2.2 The Osterwalder-Schrader theorem . . . 15
2.3 Functorial quantum field theories . . . 16

3 Rules and methods of the path integral calculus **17**
3.1 Long-time limit . . . 17
3.2 Lattice-slicing approximation . . . 19
3.3 Constrained quantisation and functional determinants . . . 21
3.4 Semi-classical approximation and perturbative expansion . . . 23
3.5 Changes of variables . . . 25
3.6 Effective field theory approach . . . 26

4 Some proofs in the symbolic calculus framework **30**
4.1 The Hellmann-Feynman theorem . . . 30
4.2 The QCD vacuum . . . 30
4.3 T-duality in string theory . . . 31

Conclusion **33**

Introduction

The path integral was first introduced by Richard Feynman [1, 2, 3] as a reformulation of quantum mechanics based on a formal analogy between the propagation of a quantum particle under Schrödinger's equation and the Brownian motion of a classical particle. Norbert Wiener had indeed constructed in 1923 a well-defined probability measure on the space of Brownian paths [4], and Feynman's insight was to replace this probabilistic weight by a phase factor involving the classical action. It was soon realised that it allowed for the quantisation of relativistic field theories where the sum over trajectories is replaced by a sum over field configurations. Path integrals thus provide a general "action-agnostic" framework for quantising classical theories.

Quantum mechanics textbooks and lectures generally do not deal with path integrals. They are typically discussed in specialised publications [3, 5, 6, 7, 8] or in Quantum Field Theory (QFT) textbooks, where they serve primarily as a computational tool for deriving perturbative expansions of propagators and scattering amplitudes [9, 10, 11, 12]. It is a pragmatic choice since most quantum field theories are rigorously defined only in the weak-coupling or perturbative regime, which is fine for example for quantum electrodynamics (where the coupling constant $\alpha_{\mathrm{QED}} = 1/137$ is genuinely small) but is a major limitation in the case of quantum chromodynamics (where the effective coupling is typically $\alpha_{\mathrm{QCD}} \sim 1$ at 200-300 MeV) or other strongly-coupled theories.

In path integral language, this is because the Wiener measure cannot be trivially generalised from continuous paths to arbitrary interacting fields (despite huge advances in mathematics over the last decades [13, 14, 15, 16]), and well-behaved functional measures are even proved not to exist in many cases of physical interest [17, 18]. Ironically, this might be the main strength of the path integral formalism: even when the rigorous quantum theory is unknown, its path integrals can still be *formally* written, and be used to establish symbolic derivations and heuristic arguments. To our surprise, although this formal calculus has spurred important advances in theoretical physics (such as quantum anomalies, the Ward-Takahashi identities, the quantisation of non-Abelian gauge fields, etc.), there seems to be no literature on path integrals as a symbolic calculus, to the exception perhaps of Skinner's QFT lecture notes [19], whose spirit we follow in this article.

Henceforth, all mathematical proofs of path integral formulae depend on specific definitions and propositions that are not applicable to all physical theories, and a proof of the systematic correspondence between the Dirac and path integral formalisms is still lacking. Accordingly, we prefer using pedagogical physical arguments to *postulate* in section 1 path integral forms for a number of quantum-mechanical quantities. Some of the limited mathematical justifications that exist for these constructions are discussed in section 2, up to the current state of the art in mathematics. Section 3 is devoted to a review of the rules and methods of symbolic path integral calculus, including time-slicing, constrained quantisation, perturbative expansions, changes of variables and effective theories, which are argued to be a general framework for most of the other methods. Mathematical comments and physical interpretations on the current understanding of these rules and methods are provided so that readers can gain intuition on their physical significance. Eventually, section 4 illustrates the power of the formalism, either on new derivations of known results (the Hellmann-Feynman theorem), or on known derivations that shed light on the physical interpretation of path integral calculus (the QCD vacuum and T-duality in string theory). The main goal of this paper is to underline how path integrals provide insights on the nature of quantum theories and how one can formally manipulate them to extract physical predictions.

1 Symbolic path integrals in quantum physics

Leaving aside all questions related to the rigorous definition of path integrals for now, we try in this section to develop an intuition for what path integrals are and how they can be handled in practice. We introduce the path integral as a formal sum over histories, derive its properties and build the functional machinery to extract physical predictions.

Subsection 1.1 introduces the space of histories and motivates the path integral as a weighted sum over all possible paths of the system. The meaning of this sum is discussed in subsection 1.2 where imposing boundary conditions leads to the definition of transition amplitudes. To manipulate these objects, the minimal techniques of functional calculus are introduced in subsection 1.3. The next logical step is to look at the representation of physical observables in this formalism, which we do in subsection 1.4. subsection 1.5 exemplifies the concepts introduced on a number of usual types of configuration histories. Having talked about probability amplitudes, we eventually introduce Wick rotation in subsection 1.6 so as to make the connection with thermalised quantum statistics.

1.1 Introduction to symbolic path integrals

In order to describe a classical physical system, we typically need to define some *parameter space* $\mathcal{M}$ (with one of the parameters usually representing time) and specify some kind of local configuration space $\mathcal{N}$ (usually called the *target space*).

The state of the system at all times is given by a map $\phi_c : \mathcal{M} \to \mathcal{N}$ called the (classical) **configuration history** of the system. In Newtonian mechanics for instance, $\mathcal{M} = \mathbb{R}$ is just time, $\mathcal{N}$ is exactly the configuration space ($\mathcal{N} = \mathbb{R}^3$ for a point particle in three dimensions), and the configuration histories $q(t)$ are the particles' trajectories. We will often refer to this case as the non-relativistic *worldline* theory. In classical field theories, $\mathcal{M}$ is usually the spacetime manifold, $\mathcal{N}$ is a space of tensors or spinors, and the configuration histories are the relativistic fields (such as the Dirac field and the four-potential in electrodynamics). To interpret path integrals, we will think of $\mathcal{M}$ as being spanned by a family of hypersurfaces $\mathcal{S}(t)$, representing the instantaneous space of parameters at time t. We call each hypersurface $\mathcal{S}$ a *worldslice* since it is a slice of the *worldvolume* $\mathcal{M}$ at a given time. In relativistic field theories, where $\mathcal{M}$ is a Lorentzian manifold describing spacetime, $\mathcal{S}$ is a spacelike hypersurface representing space at a given time. In the worldline theory, time is the only parameter and thus $\mathcal{S}$ is just a point.

We denote $\mathcal{C}$ the space of all configuration histories $\phi : \mathcal{M} \to \mathcal{N}$. It is typically infinite-dimensional and can be arbitrarily complicated and contain non-continuous functions, singular distributions, or have highly non-trivial topology. The equations of motion are encoded in the action functional $S : \mathcal{C} \to \mathbb{R}$ such that the classical history ϕ_c is selected among all $\phi \in \mathcal{C}$ by the stationary action principle $\delta S[\phi_c] = 0$. Boundary conditions and global topology may allow multiple solutions to this equation, and thus multiple classical histories. The space of histories $\mathcal{C}$ determines the number and nature of the system's degrees of freedom while the action functional S determines the system's physics, for example, its interactions, symmetries, time evolution, observable properties...

Assuming the action to be a local quantity, S is the integral over time of the Lagrangian L, which can be related to the Hamiltonian via a Legendre transform. In the Hamiltonian formulation, noting $\pi = \delta L/\delta\dot{\phi}$ the conjugate momentum to ϕ, and $\mathcal{H}$ the Hamiltonian density, the action writes

$$S[\phi] = \int_0^t \mathrm{d}\tau\, L = \int_0^t \mathrm{d}\tau \left(\int_{\mathcal{S}(\tau)} \mathrm{d}x\, (\pi\dot{\phi} - \mathcal{H}(\phi, \pi)) \right) \tag{1}$$

where the meaning of the product $\pi\dot{\phi}$ of course depends on the algebraic structure of $\mathcal{N}$. The action explicitly depends on the total time t during which the system evolves.

Classical theories are said to be deterministic since the classical history completely determines the outcome of any measurement made on the system. In quantum theories, however, the measurement outcomes follow a probability distribution! Morally, all possible configuration histories $\phi \in \mathcal{C}$ may occur with a probability that should be compatible with the correspondence principle: near-classical histories should contribute the most to quantum quantities. To implement that, we attribute a weight $\exp(\mathrm{i}S[\phi]/\hbar)$ to each history ϕ and postulate that quantum quantities are of the general form

$$\int_{\mathcal{C}} \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} F[\phi] \tag{2}$$

where $\mathcal{D}\phi$ is the *functional measure* which designates the sum over histories, and $F[\phi]$ is some functional representing the quantity we want to compute. Expressions of this form are called **Feynman path integrals**. By the stationary action principle, the action varies rapidly in regions far away from

the classical history (*i.e.* $\delta S \gg \hbar$) and so the complex exponential rotates rapidly in the complex plane. The contributions of nearby histories then cancel each other. Conversely, near the classical history, the action is slowly varying and contributions add up rather than cancel. We see that the complex exponential gives rise to *oscillatory* probabilistic weights that are very different from usual *positive definite* probability densities. This will become clear in the next subsection, when we explain that path integrals are not probability densities but probability amplitudes.

In the language of probability theory, each ϕ is a possible outcome in the trial with probability distribution $\exp(\mathrm{i}S[\phi]/\hbar)$, and each quantum quantity of the form of eq. (2) is some kind of expectation value of a random variable F. Now that we have given a general form to quantum quantities, we will work on giving a precise physical interpretation of these probabilities.

1.2 Transition amplitudes and boundary conditions

The classical configuration φ of a system at some time τ is the restriction of the configuration history ϕ to a given worldslice $\mathcal{S}$ in the parameter space $\mathcal{M}$. In other words, we may write $\varphi = \phi|_{\mathcal{S}}$. While ϕ describes the whole evolution of the system, φ only encodes its instantaneous configuration at some time τ.

Let us assume a classical system is prepared with an initial configuration φ_1 on a worldslice $\mathcal{S}_1$. Classically speaking, the probability of transitioning from φ_1 to a final configuration φ_2 on a later worldslice $\mathcal{S}_2$ is

$$\mathbb{P}(\varphi_1, \varphi_2) = \sum_j \mathbb{P}(\varphi_1, \varphi_j)\mathbb{P}(\varphi_j, \varphi_2) \tag{3}$$

where the $\{\varphi_j\}$ are the possible configurations of the system at some intermediary time. Our ignorance of what happens at this intermediary time induces the sum over j, all configuration histories (that is, all possible intermediary φ_j) contribute to the probability. Eq. (3) remains true in quantum mechanics if we perform a measurement of the system's configuration but do not look at the measurement result, so if we let the measurement device run on its own. If no measurement is performed, however, the configuration at the intermediary time does not have a well-defined value and then not only do all configuration histories contribute, they also interfere. As in optics, quantum mechanics accounts for interferences by defining a complex probability amplitude $\psi(\varphi_1, \varphi_2)$ such that

$$\mathbb{P}(\varphi_1, \varphi_2) = |\psi(\varphi_1, \varphi_2)|^2 \tag{4}$$

and by replacing eq. (3) by the more restrictive rule

$$\psi(\varphi_1, \varphi_2) = \sum_j \psi(\varphi_1, \varphi_j)\psi(\varphi_j, \varphi_2) \tag{5}$$

When squared, this equation indeed generates interference terms of the general form

$$\psi^*(\varphi_1, \varphi_j)\psi^*(\varphi_j, \varphi_2)\psi(\varphi_1, \varphi_k)\psi(\varphi_k, \varphi_2), \quad j \neq k \tag{6}$$

that cancel only when the intermediary measurement is performed because either φ_j or φ_k (or both) is zero.

Path integrals generalise this reasoning to the continuum limit: all configuration histories must contribute *and* interfere at all times. The probability of transitioning from a configuration φ_1 on $\mathcal{S}_1$ to a configuration φ_2 on $\mathcal{S}_2$ is thus the square modulus of the probability amplitude

$$\psi(\varphi_1, \varphi_2) = \langle \varphi_2 | \mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar} | \varphi_1 \rangle = \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} \tag{7}$$

with Hamiltonian operator $\hat{H}$, where each path is assumed to contribute with weight $\exp(\mathrm{i}S[\phi]/\hbar)$ because of the discussion of the previous section. $\mathcal{S}_1$ and $\mathcal{S}_2$ are two worldslices separated by time t, understood as the past and future boundaries of $\mathcal{M}$. Path integrals of the general form

$$\int_{\phi|_{\partial\mathcal{M}}=\varphi} \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S/\hbar} \tag{8}$$

are called **transition amplitudes** ($\phi|_{\partial\mathcal{M}}$ denotes the restriction of ϕ to $\partial\mathcal{M}$, the boundary of $\mathcal{M}$). In the ordinary worldline theory for example:

$$\langle q_2 | \mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar} | q_1 \rangle = \int_{q(0)=q_1}^{q(t)=q_2} \mathcal{D}q \; \mathrm{e}^{\mathrm{i}S/\hbar} \tag{9}$$

is the probability amplitude for a particle to transition from the initial state q_1 to the final state q_2 in time t, as exemplified by fig. 1.

To fix ideas, consider the simple case of the non-relativistic worldline theory, namely the propagation of a free particle of mass m, for which $\mathcal{M} = [0; t]$ is time. The action is

$$S[q] = \int_0^t \mathrm{d}\tau \, \frac{m}{2} \dot{q}^2(\tau) \tag{10}$$

The boundary $\partial\mathcal{M} = \{0\} \sqcup \{t\}$ consists of two instants, which are indeed 0-dimensional. In this example, the integral is Gaussian and the transition amplitude could be evaluated to be

$$\langle q_2 | \mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar} | q_1 \rangle = \int_{q(0)=q_1}^{q(t)=q_2} \mathcal{D}q \, \mathrm{e}^{\mathrm{i}S/\hbar} = \mathrm{e}^{-\mathrm{i}\pi/4} \sqrt{\frac{m}{2\pi\hbar t}} \exp\left(\frac{\mathrm{i}m}{2\hbar t} |q_2 - q_1|^2 \right) \tag{11}$$

Since we are only interested in formal computations and not in the evaluation of path integrals, we do not detail the proof of this formula (see however ref. [6] for a precise derivation). Note that the exponent is exactly the classical action evaluated on the straight line path defined by $q_c(\tau) = q_1 + \frac{\tau}{t}(q_2 - q_1)$, that is, the solution to the classical equations of motion. The prefactor decreasing in $t^{-1/2}$ is characteristic of the diffusion of the wave packet in time, which decreases the probability for the particle to arrive at a specific point q_2.

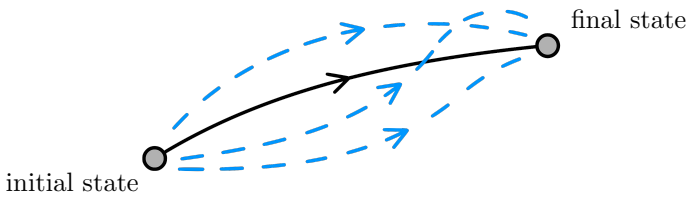

Figure 1: To travel from an initial to a final state, a particle can take several paths. The solid black line represents the classical path, it is the one dominating the sum over trajectories, while blue dashed lines are quantum-mechanical paths representing fluctuations. The path integral is then a "sum over all possible trajectories".

Back now to symbolic properties. Injecting the transition amplitudes of eq. (8) into eq. (5) yields the so-called **sewing law** of path integrals

$$\begin{aligned} \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} &= \int \mathcal{D}\varphi' \int_{\phi_1|_{\mathcal{S}_1}=\varphi_1}^{\phi_1|_{\mathcal{S}'}=\varphi'} \mathcal{D}\phi_1 \, \mathrm{e}^{\mathrm{i}S[\phi_1]/\hbar} \int_{\phi_2|_{\mathcal{S}'}=\varphi'}^{\phi_2|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi_2 \, \mathrm{e}^{\mathrm{i}S[\phi_2]/\hbar} \\ &= \int \mathcal{D}\varphi' \int_{\substack{\phi|_{\mathcal{S}_1}=\varphi_1 \\ \phi|_{\mathcal{S}'}=\varphi'}}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} \end{aligned} \tag{12}$$

where $\mathcal{S}'$ denotes some intermediary worldslice in $\mathcal{M}$. The sewing law generalises the Chasles relation of standard integrals and coincides with the Chapman-Kolmogorov-Smoluchowski equation for the non-relativistic worldline theory (see the discussion on the Feynman-Kac formula in subsection 2.1).

Of course a quantum system cannot rigorously be prepared in a *classical* configuration φ_1 on the initial worldslice $\mathcal{S}_1$. Each classical configuration can rather be obtained with a certain probability amplitude $\Psi[\varphi_1, \mathcal{S}_1]$ called the **wavefunctional**. The wavefunctional $\Psi[\varphi_2, \mathcal{S}_2]$ on the final worldslice $\mathcal{S}_2$ (which becomes the usual wavefunction $\Psi(q_2, t_2)$ at final time t_2 in the non-relativistic worldline theory) is then obtained by the sewing law, which amounts to composing the initial wavefunctional with the transition amplitude

$$\Psi[\varphi_2, \mathcal{S}_2] = \int \mathcal{D}\varphi_1 \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} \, \Psi[\varphi_1, \mathcal{S}_1] = \int^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} \, \Psi[\phi|_{\mathcal{S}_1}, \mathcal{S}_1] \tag{13}$$

Finally, because imposing boundary conditions just restricts the space of histories, integrating over all possible boundary conditions restores the full path integral:

$$\int \mathcal{D}\varphi \int_{\phi|_{\partial\mathcal{M}}=\varphi} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} = \int_{\mathcal{C}} \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S/\hbar} \tag{14}$$

1.3 Notions of functional calculus

In the previous section, we expressed quantum amplitudes as integrals over the space of histories, and since the whole idea of path integrals is to work with these functionals rather than ordinary functions, we need a few notions of functional calculus [20, 21]. Let us introduce them briefly with a simple guiding idea: functional calculus is obtained formally by extension of ordinary multivariable calculus from finite to infinite-dimensional spaces.

Functions f defined over any finite d-dimensional space depend on discrete variables x_i with $i = 1, \ldots, d$, so differentiation and integral involve sums over i. Functionals $F[\phi]$ defined over infinite-dimensional spaces depend on $\phi(t)$ where i is replaced by a continuous variable $-\infty < t < +\infty$. For our purposes, we go from finite to infinite dimensions by formally replacing

$$\sum_i \quad \to \quad \int \mathrm{d}t \tag{15}$$

The *differential* of a functional, denoted δ, is defined as usual as the first-order term in the Taylor expansion $\delta F[\phi] = F[\phi + \delta\phi] - F[\phi]$ by analogy with the finite-dimensional case

$$\mathrm{d}f = \sum_i \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i \quad \to \quad \delta F = \int \mathrm{d}t\, \frac{\delta F}{\delta\phi(t)}\delta\phi(t) \tag{16}$$

The *functional derivative* $\delta F/\delta\phi(t)$ generalises the partial derivative to functionals. Therefore, a functional derivative explicitly depends on a point t just as a partial derivative explicitly depends on a direction in space. The Leibniz rule also holds:

$$\frac{\partial(fg)}{\partial x_i} = \frac{\partial f}{\partial x_i}g + f\frac{\partial g}{\partial x_i} \quad \to \quad \frac{\delta(FG)}{\delta\phi(t)} = \frac{\delta F}{\delta\phi(t)}G + F\frac{\delta G}{\delta\phi(t)} \tag{17}$$

and the chain rule becomes:

$$\frac{\partial(f\circ g)}{\partial x_i} = \frac{\partial f}{\partial g}\frac{\partial g}{\partial x_i} \quad \to \quad \frac{\delta(f\circ G)}{\delta\phi(t)} = \frac{\partial f}{\partial G}\frac{\delta G}{\delta\phi(t)} \tag{18}$$

Note that we used a scalar function f in the last identity: general composition of two functionals is typically not defined since a functional maps a function to a scalar. If ϕ is a scalar function (that is, if $\mathcal{N}$ is a field of scalars), then $\phi(t)$ can be treated as the functional associating the *value* $\phi(t)$ to the function ϕ, for some fixed point t (it is the Dirac distribution centred on t applied on the function ϕ: $\langle\delta_t, \phi\rangle$). More generally, we note $f(\phi(t))$ the functional associating the value $f(\phi(t))$ to the function ϕ, for some fixed point t. The derivatives of these functionals can be evaluated by

$$\frac{\partial x_j}{\partial x_k} = \delta_{jk} \quad \to \quad \frac{\delta\phi(t)}{\delta\phi(t')} = \delta(t - t') \tag{19}$$

and the expected exchange rule holds:

$$\frac{\delta}{\delta\phi(t)}\int \mathrm{d}\tau\, f(\phi(\tau)) = \int \mathrm{d}\tau\, \frac{\delta f(\phi(\tau))}{\delta\phi(t)} \tag{20}$$

Note that a "standard" derivative $\partial/\partial\lambda$ for an arbitrary parameter λ is a special case of functional derivative, and thus standard derivatives can also be exchanged with functional integrals.

Finally, by Stokes' theorem, assuming F to be compactly supported on $\mathcal{C}$,

$$\int_{\mathcal{C}} \mathcal{D}\phi\, \delta F[\phi] = \int_{\partial\mathcal{C}} F = 0 \tag{21}$$

since functional spaces are usually without boundary. This leads in particular to

$$\int \mathcal{D}\phi\, \frac{\delta F}{\delta\phi(t)} = 0 \tag{22}$$

Finally, by replacing F by a product FG in this equation and applying Leibniz' rule, we get the *functional integration by parts* formula

$$\int \mathcal{D}\phi\, \frac{\delta F}{\delta\phi(t)}G[\phi] = -\int \mathcal{D}\phi\, F[\phi]\frac{\delta G}{\delta\phi(t)} \tag{23}$$

1.4 Observable functionals and Schwinger-Dyson equations

Having established the path integral as a framework for computing probability amplitudes, we now turn to the question of how physical observables are represented.

In classical physics, an **observable** $O(\phi, \pi, t)$ is a function of the generalised coordinates, their conjugate momenta, and possibly even time[1]. Alternatively, if one knows the complete configuration history ϕ of the system, then one also knows $\dot{\phi}$ and thus $\pi = \delta L/\delta\dot{\phi}$ (if L is the classical Lagrangian) at all times. Therefore, any observable can also be seen as a functional $O[\phi]$ of the configuration *history*. In this sense, the observable at time τ corresponds to a functional $O_\tau[\phi]$, which depends on the history up to τ and which is different from the functional $O_{\tau'}$ measuring the same observable at τ'. Hence, a classical observable really corresponds to a family of functionals $O_\tau[\phi]$ indexed by τ.

If the observable $O(\phi, \pi)$ has no explicit time dependence, then $O_\tau[\phi]$ only depends on the configuration history in an infinitesimal period $2\delta t$ around τ, since knowing ϕ between $\tau - \delta t$ and $\tau + \delta t$ is sufficient to reconstruct $\dot{\phi}$ and thus π. We say that it depends only on $\phi(\tau)$ in the vicinity of τ and write $O_\tau[\phi] = O(\tau)$ if the configuration variable is obvious. The classical observable functional is then formally related to the quantum operator by

$$\langle\varphi_2|\mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar}\hat{O}(\tau)|\varphi_1\rangle = \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}O(\tau) \tag{24}$$

where the time dependence of the operator $\hat{O}(\tau)$ arises from working in the Heisenberg picture. This immediately implies the *linearity in the integrand* of the path integral

$$\int_{\mathcal{C}} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}O_1 + \int_{\mathcal{C}} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}O_2 = \int_{\mathcal{C}} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}(O_1 + O_2) \tag{25}$$

More generally, since the action exponential is seen as a probabilistic weight, path integrals of the form

$$\langle O\rangle_{\mathcal{C}} = \frac{1}{Z}\int_{\mathcal{C}} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}O[\phi] \tag{26}$$

are interpreted as **expectation values** against a choice $\mathcal{C}$ of configuration histories. The normalisation of the expectation is ensured by the **partition function**

$$Z = \int_{\mathcal{C}} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar} \tag{27}$$

Normalisation has a special importance for path integrals since, even when they are well-defined and computable, their convergence towards a finite value is rarely guaranteed. In many cases of physical interest, the divergences of the integral $\int \mathcal{D}\phi\ \exp(\mathrm{i}S/\hbar)O$ are exactly compensated by the divergences of Z, so that the expectation value is finite [10, 22].

The notation $\langle O\rangle_{\mathcal{C}}$ should not be confused with the operator expectation value $\langle\psi|\hat{O}|\psi\rangle\,/\,\langle\psi|\psi\rangle$ since not all expectation values in the sense of path integrals are expectation values against a quantum state. We will see in subsection 3.1 how expectation values against stationary states can be obtained from path integral expectation values by the long-time principle.

Explicitly constructing a Hilbert space operator for every observable is not straightforward and is in fact the central problem of canonical quantisation. Following the work of Dirac [23], one promotes the canonical variables to operators $\hat{\phi}, \hat{\pi}$ satisfying canonical commutation relations. These correspond to the functionals ϕ and $\pi = \delta L/\delta\dot{\phi}$. Operators that are polynomials in $\hat{\phi}$ or in $\hat{\pi}$ then correspond to polynomials in the functionals ϕ and π, and the prescription is easily extended linearly. However, difficulties start to appear when we consider products of $\hat{\phi}$ and $\hat{\pi}$ since these operators do not commute whereas the corresponding functionals usually commute (they are scalar-valued). A prescription must be chosen, usually between

$$\begin{aligned} \text{Normal ordering:} \quad & \phi\pi \mapsto \hat{\phi}\hat{\pi} \\ \text{Antinormal ordering:} \quad & \phi\pi \mapsto \hat{\pi}\hat{\phi} \\ \text{Weyl ordering:} \quad & \phi\pi \mapsto (\hat{\phi}\hat{\pi} + \hat{\pi}\hat{\phi})/2 \end{aligned} \tag{28}$$

It is a sad reality that these prescriptions are inequivalent and that no universal principle decides between them. Even worse, a theorem by Groenewold states that no ordering prescription can save the correspondence from ambiguities [24, 25]. Path integral quantisation formally dodges this

[1] To be precise, if ϕ is a field, then O is a *functional* of $\phi|_{\mathcal{S}}$ for some instantaneous worldslice $\mathcal{S}$. This precision is not important for the present discussion.

difficulty: rather than constructing quantum operators explicitly, it expresses quantum quantities in terms of classical functionals! Let us mention for completeness that the operator ordering problem is even more complicated in curved spacetime. When the metric depends on position, we also have to decide how to order it relative to momentum operators, and different choices lead to physically different Schrödinger equations. A systematic resolution of this problem in path integrals can be found in ref. [26].

In general, the product of two operators $\hat{O}_1\hat{O}_2$ is not equal to $\hat{O}_2\hat{O}_1$ whereas the product of two functionals $O_1[\phi]O_2[\phi]$ does commute, so the path integral of a product of observables must provide a "natural" ordering, in the form of the **time-ordered product**. It means the matrix elements precisely correspond to

$$\langle O_2(\tau_2)O_1(\tau_1)\rangle_{\mathcal{C}} = \int_{\phi|_{S_1}=\varphi_1}^{\phi|_{S_2}=\varphi_2} \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar} O_2(\tau_2)O_1(\tau_1) = \langle\varphi_2|\mathcal{T}\Big[\mathrm{e}^{\mathrm{i}t\hat{H}/\hbar}\hat{O}_2(\tau_2)\hat{O}_1(\tau_1)\Big]|\varphi_1\rangle \tag{29}$$

where $\mathcal{T}$ is the time-ordering symbol defined such that

$$\mathcal{T}\Big[\hat{O}_2(\tau_2)\hat{O}_1(\tau_1)\Big] = \mathcal{T}\Big[\hat{O}_1(\tau_1)\hat{O}_2(\tau_2)\Big] = \hat{O}_2(\tau_2)\hat{O}_1(\tau_1) \tag{30}$$

for $\tau_1 < \tau_2$ [2].

As an important application of the properties of observables, let us integrate by part some functional derivative [21]

$$\int \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}\frac{\delta O}{\delta\phi(x)} = -\int \mathcal{D}\phi\ O\frac{\delta \mathrm{e}^{\mathrm{i}S/\hbar}}{\delta\phi(x)} = -\frac{\mathrm{i}}{\hbar}\int \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}O\frac{\delta S[\phi]}{\delta\phi(x)} \tag{31}$$

which immediately yields:

Theorem (Schwinger-Dyson equations)**.** *For any $x \in \mathcal{M}$, any space of histories $\mathcal{C}$ where boundaries vanish, any action functional S and any functional O:*

$$\left\langle O\frac{\delta S}{\delta\phi(x)}\right\rangle_{\mathcal{C}} = \mathrm{i}\hbar\left\langle\frac{\delta O}{\delta\phi(x)}\right\rangle_{\mathcal{C}} \tag{32}$$

This is a fundamental result whose non-relativistic expression dates back to Feynman [2] and was later extended by Schwinger and Dyson [27, 28]. For the sake of completeness, note the important hypothesis that $\mathcal{C}$ should be *without boundaries* (or that the functionals should cancel at the boundaries). This is typically true if the parameter space $\mathcal{M}$ is chosen to be without boundaries, that is, if we take the past and future worldslices to be at infinity. This is the so-called long-time limit that yields stationary state expectation values, as will be discussed in section 3.1.

Direct applications to various functionals can be found in ref. [29]. In particular, taking $O = 1$ immediately yields Ehrenfest's relations

$$\left\langle\frac{\delta S}{\delta\phi(x)}\right\rangle_{\mathcal{C}} = 0 \tag{33}$$

demonstrating that the Euler-Lagrange equations of motion hold in expectation. There is one especially illuminating application of this equation that we want to underline. We know that for any local action S and Lagrangian density $\mathcal{L}(\phi, \partial_\mu\phi)$:

$$\frac{\delta S}{\delta\phi(x,t)} = \frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi} \tag{34}$$

where x, t respectively denote worldslice and time coordinates. Taking $O = \phi(x_1, \tau_1)$ and the derivative with respect to $\phi(x_2, \tau_2)$ yields

$$\int \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}\phi(x_1,\tau_1)\left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right) = \mathrm{i}\hbar\ \delta^{d-1}(x_1 - x_2)\delta(\tau_1 - \tau_2) \tag{35}$$

If we integrate the l.h.s. on all x_2, and on τ_2 between $\tau_1 - \varepsilon$ and $\tau_1 + \varepsilon$, we get

$$\frac{1}{Z}\int \mathrm{d}x_2 \int \mathcal{D}\phi\ \mathrm{e}^{\mathrm{i}S/\hbar}\phi(x_1,\tau_1)\left(\int_{\tau_1-\varepsilon}^{\tau_1+\varepsilon}\frac{\partial\mathcal{L}(x_2,\tau_2)}{\partial\phi}\,\mathrm{d}\tau_2 + \frac{\partial\mathcal{L}(x_2,\tau_1-\varepsilon)}{\partial\dot{\phi}} - \frac{\partial\mathcal{L}(x_2,\tau_1+\varepsilon)}{\partial\dot{\phi}}\right) \tag{36}$$

because the histories $\phi \in \mathcal{C}$ are taken to vanish at spatial infinity, so the boundary terms $\partial_i(\partial\mathcal{L}/\partial\partial_i\phi)$ vanish. This expression simplifies: first in the limit $\varepsilon \to 0$ the integral of $\partial\mathcal{L}/\partial\phi$ cancels by continuity of the antiderivative, second $\partial\mathcal{L}/\partial\dot{\phi} = \pi$, and finally we can get rid of the integration on x_2 by introducing a function of x_2 with arbitrary compact support before integrating and using the fundamental lemma of the calculus of variations. Then, we get

$$\frac{1}{Z}\int \mathcal{D}\phi\ \left(\phi(x_1,\tau_1)\pi(x_2,\tau_1^-) - \phi(x_1,\tau_1)\pi(x_2,\tau_1^+)\right) = \mathrm{i}\hbar\ \delta^{d-1}(x_1 - x_2) \tag{37}$$

Because of time ordering, it is exactly the canonical commutation relation:

$$\left\langle \left[\hat{\phi}(x_1), \hat{\pi}(x_2)\right] \right\rangle_{\mathrm{op}} = \mathrm{i}\hbar\ \delta^{d-1}(x_1 - x_2) \tag{38}$$

where $\langle \hat{O} \rangle_{\mathrm{op}}$ is the expectation value in the sense of operators (we anticipate on the connection with operator expectation values made in subsection 3.1).

Time limits and time derivatives. This derivation explicitly shows that, even formally, path integrals *do not commute with limits*, or else the l.h.s. of eq. (37) would vanish by continuity of the canonical variables! The discontinuity of the time-ordered product at equal times thus enforces the fundamental non-commutativity of quantum theories. Furthermore, functionals in formal path integrals should never be directly evaluated at equal times, except if their commutator (computed as in eq. (37)) is shown to vanish.

The non-commutation of time limits and path integrals implies a counter-intuitive result of functional integration: *the histories that contribute to the path integral are nowhere differentiable* (see subsection 2.1 for a discussion, and refs. [29, 30]). This is deeply disturbing: we postulated path integral forms for quantum-mechanical quantities because the (typically infinitely differentiable) classical path was apparently the main contributor to the path integral, at least in the classical limit. Imposing the non-commutativity of quantum mechanics leads to refining our initial postulate: the dominant contribution to the path integral does not come from the classical history itself but from *nowhere differentiable histories in the neighbourhood of the classical history.*

Under the exchange rule (20), however, time derivatives do commute with path integrals. This can be understood in a simple way: time is just a *parameter* in $\mathcal{M}$, and the path integral measure should be invariant under reparametrisation (typically, path integrals should be Lorentz-covariant quantities in relativistic theories). Such an invariance forbids the measure to depend explicitly on a choice of time parameter, and thus the path integral should commute with time derivatives. For this reason, the symbol $\mathcal{T}$ appearing in eq. (29) is the so-called *covariant* time-ordering symbol, which commutes with time derivatives [31], and not the naive time-ordering symbol $\mathcal{T}_{\mathrm{naive}}\hat{O}_1(\tau_1)\hat{O}_2(\tau_2)$ defined as $\hat{O}_1(\tau_1)\hat{O}_2(\tau_2)\Theta(\tau_1 - \tau_2) + \hat{O}_2(\tau_2)\hat{O}_1(\tau_1)\Theta(\tau_2 - \tau_1)$ where the Heaviside functions of time will of course yield additional terms upon differentiation.

Generalising the commutation relations. Of course, the derivation of eq. (38) is only valid if $\phi(x_1,\tau_1)$ is a functional, that is, if it has scalar values. In general, ϕ is a function from $\mathcal{M}$ to $\mathcal{N}$, where $\mathcal{N}$ may *a priori* be a spinor or tensor space. We are saved by considering components ϕ^Λ of ϕ, where Λ is a very general multi-index possibly containing spinor indices, tensor indices, complex conjugation,... For example, if ϕ is a complex scalar field, then we may write $\phi^\Lambda = (\phi, \phi^*)$, if ϕ is a tensor product of a real endomorphism and a Dirac spinor, then we may write $\phi^\Lambda = \phi^\mu{}_\nu{}^{\hat{\alpha}}$. Lowering the index then means taking whatever canonical duality is defined on $\mathcal{N}$ (for example Fréchet-Riesz duality or Dirac conjugation), and contracting indices means applying whatever canonical pairing is defined on $\mathcal{N}$ (such as inner product or Lorentzian product). With this notation, ϕ^Λ is a real or complex number, and $\phi^\Lambda(x_1,\tau_1)$, for some fixed $(x_1,\tau_1) \in \mathcal{M}$, is indeed a functional. The Schwinger-Dyson equations (31) remain correct since ϕ need not be a functional for the functional derivative to be well-defined (any function works just fine), but they can be refined:

$$\left\langle O\frac{\delta S}{\delta\phi^\Lambda(x)} \right\rangle_{\mathcal{C}} = \mathrm{i}\hbar \left\langle \frac{\delta O}{\delta\phi^\Lambda(x)} \right\rangle_{\mathcal{C}} \tag{39}$$

The proof of the canonical commutation relations, however, must be adapted because we must take $O = \phi^\Sigma(x_1,\tau_1)$, and with the same steps we obtain

$$\frac{1}{Z}\int \mathcal{D}\phi\ \left(\phi^\Sigma(x_1,\tau_1)\pi_\Lambda(x_2,\tau_1^-) - \phi^\Sigma(x_1,\tau_1)\pi_\Lambda(x_2,\tau_1^+)\right) = \mathrm{i}\hbar\ \delta^{d-1}(x_1 - x_2)\ \delta^\Sigma{}_\Lambda \tag{40}$$

and the canonical commutation relations follow.

The situation worsens if we choose the components ϕ^Λ of the target space elements ϕ to be anticommuting numbers (a-numbers, or Grassmann variables, or Grassmann-odd supernumbers) rather than commuting numbers (c-numbers, or Grassmann-even supernumbers). Here again, the Schwinger-Dyson equations do not suffer from this since the action is Grassmann-even in any case. For the same reason, and as observables are generally Grassmann-even too, most of the formal calculus is unaffected by this choice of target space. Even eq. (40) remains correct since we do not force ϕ^Σ to commute with π_Λ to obtain it. But as the second term in the l.h.s. is submitted to time-ordering to interpret it in the operator formalism, a minus sign appears because the variables ϕ^Σ, π_Λ now anticommute. In this case, canonical *anticommutation* relations follow:

$$\left\langle \left\{ \hat{\phi}^\Sigma(x_1), \hat{\pi}_\Lambda(x_2) \right\} \right\rangle_{\text{op}} = \mathrm{i}\hbar\, \delta^{d-1}(x_1 - x_2)\, \delta^\Sigma{}_\Lambda \tag{41}$$

Correlation functions. To complete this section, let us mention that one of the most important classes of observables are the n-point correlation functions, obtained by inserting field values at distinct spacetime points:

$$\langle \phi^{\Lambda_1}(x_1) \dots \phi^{\Lambda_n}(x_n) \rangle_{\mathcal{C}} = \frac{1}{Z} \int_{\mathcal{C}} \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar} \phi^{\Lambda_1}(x_1) \dots \phi^{\Lambda_n}(x_n) \tag{42}$$

because they encode the full quantum dynamics of the theory. All scattering amplitudes and response functions can be derived from them. These correlation functions can be generated by introducing an arbitrary classical source function $J : \mathcal{M} \to \mathcal{N}$ coupled linearly to the field and defining the **generating functional**

$$Z[J] = \int \mathcal{D}\phi\, \exp\left(\frac{\mathrm{i}}{\hbar} S[\phi] + \frac{\mathrm{i}}{\hbar} \int_{\mathcal{M}} J_\Sigma(x) \phi^\Sigma(x)\, \mathrm{d}x \right) \tag{43}$$

so that all n-point correlation functions can be obtained as *functional derivatives* of the generating functional:

$$\langle \phi^{\Lambda_1}(x_1) \dots \phi^{\Lambda_n}(x_n) \rangle_{\mathcal{C}} = \frac{1}{Z[0]} \left(\frac{\hbar}{\mathrm{i}} \right)^n \frac{\delta^n Z[J]}{\delta J_{\Lambda_1}(x_1) \dots \delta J_{\Lambda_n}(x_n)} \bigg|_{J=0} \tag{44}$$

How do these observables behave with respect to the Schwinger-Dyson equations? Setting $O = \phi^{\Lambda_1}(x_1) \dots \phi^{\Lambda_n}(x_n)$ generates a hierarchy of relations between n- and $(n+1)$-point correlation functions in the sense that

$$\left\langle \frac{\delta S}{\delta \phi^\Lambda(x)} \phi^{\Lambda_1}(x_1) \dots \phi^{\Lambda_n}(x_n) \right\rangle_{\mathcal{C}} = \mathrm{i}\hbar \sum_{k=1}^{n} \delta^{(d)}(x - x_k) \delta^\Lambda{}_{\Lambda_k} \left\langle \phi^{\Lambda_1}(x_1) \cdots \widehat{\phi^{\Lambda_k}(x_k)} \dots \phi^{\Lambda_n}(x_n) \right\rangle_{\mathcal{C}} \tag{45}$$

where the hat is a symbol used for omission. Importantly, the Schwinger-Dyson equations remain meaningful even when the path integral measure is not rigorously defined, so they can be taken as a powerful defining property of the quantum theory in the absence of a well-defined measure. We thus have seen that path integrals provide an elegant way of computing correlation functions and have identified some key properties that will be useful for the discussions to come.

1.5 Various examples of target spaces

As may be seen from the previous section, the choice of a target space $\mathcal{N}$ has an impact on at least certain path integrals (here, the correlation functions). Fundamentally, only the space of histories $\mathcal{C}$ has a real physical significance since it is the domain of the path integral. The target space, as well as the parameter space $\mathcal{M}$, are only useful in the construction of $\mathcal{C}$. As we will see in subsection 3.5, we can perform changes of variables inside path integrals that may completely transform $\mathcal{M}$ and $\mathcal{N}$, as long as it leaves $\mathcal{C}$ invariant up to diffeomorphism. In practice, however, we want to make *physically-motivated* choices of target spaces so as to justify the construction of $\mathcal{C}$. In relativistic field theory, for instance, the possible target spaces are classified by the irreducible unitary representations of the Poincaré group in order to ensure the Poincaré-invariance of the quantum theory.

The choice of target space does not affect most of the formal calculus since the histories generally only appear as bound variables, but the physical theories we ultimately describe involve fields of different natures. We describe in this section various typical examples of target spaces.

A *scalar field* $\phi : \mathcal{M} \to \mathbb{R}$ or $\mathbb{C}$ is the simplest type of field as it assigns a single value to each spacetime point. It has no internal degree of freedom nor spacetime index, so it transforms

trivially under spacetime and internal symmetries. In this case, the space of histories is simply made of functions, and the path integral measure $\mathcal{D}\phi$ is understood as a (fictitious, as discussed in subsection 2.1) infinite-dimensional Lebesgue measure. The typical action for real scalar fields is

$$S[\phi] = \int_{\mathcal{M}} \mathrm{d}^d x \left[\frac{1}{2}(\partial_\mu \phi)^2 - V(\phi)\right] \tag{46}$$

with kinetic and potential terms. Scalar fields typically belong to the spin-0 representation space of the Poincaré group. By comparison with the real case, the complex scalar field carries an additional U(1) symmetry that gives rise to a conserved current under Noether's theorem.

A *gauge field* is a one-form $A_\mu = A^a_\mu t^a$ taking values in the Lie algebra $\mathfrak{g}$ of a gauge group G with generators t^a. Canonical examples are Quantum Electrodynamics (QED) with $G = \mathrm{U}(1)$ and a single real component A_μ, and Quantum Chromodynamics (QCD) with $G = \mathrm{SU}(3)$ and eight a components A^a_μ. The classical action is built from the field strength $F_{\mu\nu} = \partial_{[\mu} A_{\nu]} - \mathrm{i}[A_\mu, A_\nu]$ (this is the gauge-covariant generalisation of the electromagnetic tensor, for which the Lie bracket vanishes), giving

$$S[\phi] = -\frac{1}{4\mu_0} \int_{\mathcal{M}} \mathrm{d}^d x \ \mathrm{tr}\{F_{\mu\nu} F^{\mu\nu}\} \tag{47}$$

The space of histories is made complicated by the presence of gauge redundancies: the classical state is determined by the field strength (such as the electric and magnetic fields in QED), which is invariant under local gauge transformations

$$A_\mu(x) \to U(x) A_\mu(x) U^{-1}(x) + \mathrm{i} U(x) \partial_\mu U^{-1}(x), \qquad U(x) \in G \tag{48}$$

This means that many different configurations of the gauge field represent the same physical state, so naive, Lebesgue-like integration over A_μ overcounts configurations and leads to irrecuperable divergences in path integrals. The correct (but still fictitious) integration measure is a sum over gauge-inequivalent classes of gauge fields. As explained in subsection 3.3, this is usually handled by imposing a gauge-fixing condition [19].

Describing *fermion fields* is a bit more intricate [32]. Starting with the representation of the Poincaré algebra, we can derive that such particles are described by Dirac spinor fields ψ taking values in $\mathbb{C}^2 \oplus \mathbb{C}^2$. It is a four-component, complex-valued field, and we define its Dirac conjugate $\overline{\psi} = \psi^\dagger \gamma^0$ using the Dirac matrices γ^μ. These matrices close a Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. The classical action is built from the Dirac operator

$$S[\psi, \overline{\psi}] = \int_{\mathcal{M}} \mathrm{d}^d x \ \overline{\psi}(\mathrm{i}\hbar c \gamma^\mu \partial_\mu - mc^2)\psi \tag{49}$$

where m is the mass of the fermion. The crucial difference is that fermions obey canonical anticommutation relations, which may be enforced by taking the path integration variables to be anticommuting, as was observed in subsection 1.4 (we have $\psi^{\hat{\alpha}}(x)\psi^{\hat{\beta}}(y) = -\psi^{\hat{\beta}}(y)\psi^{\hat{\alpha}}(x)$). This is merely a choice because quantising with commutation relations produces an energy spectrum that is not bounded from below, making it impossible to extract finite energies. In some cases however, unobservable spin-half particles, such as Faddeev-Popov ghosts [33], have no physical energy spectrum, and thus may be Grassmann-even (and thus have canonical commutation relations). The fictitious Lebesgue measure $\mathcal{D}\psi \ \mathcal{D}\overline{\psi}$ is considerably simplified by the anticommutation prescription, which kills all squares, for instance, and reduces to a *well-defined* infinite-dimensional Berezin integral over a Grassmann algebra. Note that the Grassmann-odd fields ψ and $\overline{\psi}$ are treated as independent.

In each case, the path integral retains the same symbolic form, that of a sum over configuration histories weighted by a complex exponential, what changes is the nature of the fields being summed over and thus the rules we must follow to manipulate them. The symbolic approach we use in this article is largely insensitive to these distinctions precisely because it operates at the level of the action and measure rather than that of the field content.

1.6 Wick rotation and statistical physics

Understanding the path integral formalism as a probabilistic theory makes quantum theory really reminiscent of thermalised statistics. In both cases, we are left to sum over large numbers of possible configurations, weighted by a proper exponential factor. To really make this connection explicit, we will take a small detour by the canonical ensemble before re-entering the world of quantum path integrals. We will later see in subsection 2.2 how to rigorously bridge these two visions.

In statistical mechanics, the canonical density matrix of a system at temperature T is $\exp\left(-\beta\hat{H}\right)$, where $\beta = 1/(k_B T)$ is defined using the Boltzmann constant k_B. Then, per eq. (7):

$$\langle\varphi_2|\mathrm{e}^{-\beta\hat{H}}|\varphi_1\rangle = \langle\varphi_2|\mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar}|\varphi_1\rangle\Big|_{t=-\mathrm{i}\hbar\beta} = \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi\ \mathrm{e}^{-S_E/\hbar} \tag{50}$$

where

$$S_E[\phi] = \int_0^{\hbar\beta} \mathrm{d}\tau \int \mathrm{d}x\,(\pi_E\partial_\tau\phi + \mathcal{H}(\phi,\pi_E)) \tag{51}$$

is the so-called *Euclidean action* obtained by replacing t by the **imaginary time** $-\mathrm{i}\hbar\beta = -i\tau$ in the physical action S. The Euclidean canonical momentum is $\pi_E = -\mathrm{i}\pi$ and $\mathcal{H}$ is the Hamiltonian density. This "replacement" is called a **Wick rotation** because it amounts to a $\pi/2$ rotation of the time parameter in the complex plane. It turns oscillatory quantum amplitudes into exponentially damped statistical amplitudes, and so formally maps a quantum *Feynman path integral* in real time onto a statistical **Euclidean path integral** in imaginary time. Note that Wick rotation literally turns Minkowski space $\mathbb{R}^{d-1,1}$ into Euclidean space $\mathbb{R}^{d,0}$, so quantum field theories are statistical field theories in one higher space dimension. The canonical partition function then becomes

$$\begin{aligned}\mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}}\right] = \int \mathcal{D}\varphi\ \langle\varphi|\mathrm{e}^{-\beta\hat{H}}|\varphi\rangle &= \int \mathcal{D}\varphi\ \langle\varphi|\mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar}|\varphi\rangle\Big|_{t=-\mathrm{i}\hbar\beta} \\ &= \int \mathcal{D}\varphi \int_{\phi|_{\mathcal{S}_1}=\varphi}^{\phi|_{\mathcal{S}_2}=\varphi} \mathcal{D}\phi\ \mathrm{e}^{-S_E/\hbar}\end{aligned} \tag{52}$$

As this partition function is defined by integrating the Euclidean action, we will now denote it Z_E. This can be more compactly written as a path integral over closed paths in imaginary time

$$Z_E = \oint \mathcal{D}\phi\ \mathrm{e}^{-S_E/\hbar} \tag{53}$$

which is *slightly* different from the quantum partition function of eq. (27). This imposes a periodicity $\tau \to \tau + \beta$ encoding the thermal nature of the system and is the basis for building finite-temperature field theory [10]. The statistical expectation value of an observable (in the Schrödinger picture) is now of the form

$$\begin{aligned}\langle O\rangle_\beta = \frac{1}{Z_E}\,\mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}}\hat{O}\right] &= \frac{1}{Z_E}\int \mathcal{D}\varphi_1\,\mathcal{D}\varphi_2\ \langle\varphi_2|\mathrm{e}^{-\beta\hat{H}}|\varphi_1\rangle\ \langle\varphi_1|\hat{O}|\varphi_2\rangle \\ &= \frac{1}{Z_E}\int \mathcal{D}\varphi_1\,\mathcal{D}\varphi_2 \int_{\phi|_{\mathcal{S}_1}=\varphi_1}^{\phi|_{\mathcal{S}_2}=\varphi_2} \mathcal{D}\phi\ \mathrm{e}^{-S_E/\hbar}\,\langle\varphi_1|\hat{O}|\varphi_2\rangle \\ &= \frac{1}{Z_E}\int \mathcal{D}\phi\ \mathrm{e}^{-S_E/\hbar} O[\phi|_{\partial\mathcal{M}}]\end{aligned} \tag{54}$$

where $O[\phi|_{\partial\mathcal{M}}]$ is the general form of the functional associated to the observable O, which is $\langle\varphi_1|\hat{O}|\varphi_2\rangle$ in the case where $\partial\mathcal{M} = \mathcal{S}_1 \sqcup \mathcal{S}_2$ for two worldslices $\mathcal{S}_1, \mathcal{S}_2$. Similarly, thermal correlation functions are computed by inserting operators at Euclidean times τ_i.

Because real exponentials are easier to work with than complex exponentials, both theoretically and numerically, a usual trick in quantum field theory consists in performing all computations in this Euclidean framework, and, in the end, analytically continuing the resulting thermal correlation functions to real time to obtain quantum correlation functions.

We have so far introduced a great deal of formal objects and used them without caring about their meaning nor any kind of mathematical rigour. By now, questions related to the definition of path integrals and to the justification and validity of Wick rotation must be crawling in the mind of the reader, so it seems worth momentarily interrupting the discussion about formal calculus to try and shed light on the deeper roots of path integrals.

2 Mathematical approaches to path integrals

Since path integrals are typically ill-defined, it can be cumbersome trying to understand how much of path integral computations is indeed mathematically justified. This section tackles various mathematical aspects of path integral calculus and thus may be skipped at first reading. Some knowledge of measure theory, stochastic processes or category theory may be useful to fully grasp the corresponding subsections, but fear not: references are provided along the way for the interested reader.

This section is organised around three complementary perspectives on the problem. As we will see in subsection 2.1, Euclidean worldline path integrals are always well-defined because they are integrals against the *Wiener measure*. Moreover, the so-called Feynman-Kac formula rigorously identifies the solutions of the imaginary-time Schrödinger equation as Euclidean path integrals. The analytic continuation to real time is then justified in a large class of cases by the Osterwalder-Schrader theorem, addressed in subsection 2.2 in the context of axiomatic quantum field theories. In turn, this motivates in subsection 2.3 the study of Feynman integrals through their properties alone: the so-called functorial field theories abandon the very idea of functional integration to define Feynman integrals as abstract functors between categories of cobordisms and Hilbert spaces.

For the sake of brevity, other approaches are not developed here although they can be quite insightful and very interesting, namely those based on the definition of path integrals as (pro)distributions, objects that generalise the concept of (pro)measures [34, 14, 35], on white noise calculus, an extension of stochastic calculus [13], or on cohomology classes, namely in the context of the Batalin-Vilkovisky formalism (see *e.g.* ref. [36] for the connection between the BV-cochain complex and the de Rham complex on configuration space).

2.1 The Wiener measure and the Feynman-Kac formula

The mathematical theory of functional integration is rooted in the seminal work of Wiener [4] who defined a rigorous probability measure to evaluate expectation values in Brownian processes. As usual, the probability measure can be understood as an integration measure, except this time the so-called Wiener measure is a *functional* integration measure since the space of Brownian trajectories is a functional space.

First thing to note however, is that there is no "natural" measure in infinite-dimensional functional spaces, and these spaces make the discussion of volume quite tricky. The most famous result of this kind is the assertion that there is no finite non-trivial translation-invariant Borel measure on infinite-dimensional separable Banach spaces [37]. Such a measure would be natural in the sense that it would resemble the well-known Lebesgue measure of finite-dimensional $\mathbb{R}^d$ spaces: it is finite since it associates a finite volume to any bounded domain of $\mathbb{R}^d$ (which is what we want from a volume), and translation-invariant since the volume does not depend on the point around which the bounded domain is centred.

The common practice in physics is to denote $\mathcal{D}q(t)$ the (fictitious, non-existent) Lebesgue measure on the space $\{q(t)\}$ of continuous bounded paths in $\mathbb{R}^d$ with fixed final value $q_2 \in \mathbb{R}^d$ at time t (we set the initial time at 0, where paths can have any possible value in $\mathbb{R}^d$). It does not really correspond to any well-defined object, but the insight of Wiener was to abandon translation invariance and construct a real Gaussian measure on the space of continuous paths. The Wiener Gaussian measure can then be *formally* related [38] to the fictitious natural measure $\mathcal{D}q(t)$ by

$$\mathrm{d}W_{q(t)} = \exp\left(-\frac{m}{2\hbar}\int \dot{q}^2(t)\,\mathrm{d}t\right)\mathcal{D}q(t) \tag{55}$$

with m the mass of the particle. The exponential factor explicitly breaks translation invariance since adding a shift path $\delta q(t)$ to all trajectories modifies $\dot{q}$ and so the measure. However, or rather fortunately, the loss of translation invariance has a natural physical meaning since it comes from the incorporation of the particles' kinetic energy into the measure. As we will see from the Feynman-Kac formula, this is exactly how the classical action emerges in the path integral.

We will not dwell on the non-trivial mathematical construction of this measure here, since for many physical applications, knowing such a measure exists is sufficient. The well-defined Wiener measure is related to the evaluation of physical quantities through the *Feynman-Kac formula*, which is at the very basis of the whole idea of path integration [39, 40]. It states that a physical quantity satisfying a certain elliptic equation (like the diffusion equation of Brownian motion) can be written as a functional integral against the Wiener measure.

Theorem (Feynman-Kac formula)**.** *Let $\Psi(q,t)$ be a function satisfying the elliptic equation*

$$\frac{\partial\Psi(q,t)}{\partial t} = \frac{1}{\hbar}\left[\frac{\hbar^2}{2m}\Delta - V(q) + \mathrm{i}s(t)q\right]\Psi(q,t) \tag{56}$$

where V is a potential energy and $s(t)$ is a real time-dependent source term. Then

$$\Psi(q,t) = \int \mathrm{d}W_{\xi(\tau)}\,\Psi(\xi,0)\,\exp\left(\frac{\mathrm{i}}{\hbar}\int_0^t s(\tau)\xi(\tau)\,\mathrm{d}\tau\right)\exp\left(-\frac{1}{\hbar}\int_0^t V(\xi)\,\mathrm{d}\tau\right) \tag{57}$$

where the Wiener paths $\xi(\tau)$ end at $\xi(t) = q$, and with $\Psi(\xi,0)$ a choice of initial conditions.

Rewriting this formula in terms of the fictitious Lebesgue measure yields

$$\Psi(q,t) = \int \mathcal{D}\xi(\tau)\ \Psi(\xi,0)\ \exp\left(\frac{\mathrm{i}}{\hbar}\int_0^t s(\tau)\xi(\tau)\,\mathrm{d}\tau\right)\exp\left(-\frac{m}{2\hbar}\int_0^t \dot{\xi}^2(\tau)\,\mathrm{d}\tau - \frac{1}{\hbar}\int_0^t V(\xi)\,\mathrm{d}\tau\right) \tag{58}$$

and we see that the exponentiated quantity

$$\frac{m}{2}\int_0^t \dot{\xi}^2(\tau)\,\mathrm{d}\tau + \int_0^t V(\xi)\,\mathrm{d}\tau \tag{59}$$

is exactly the Euclidean version of the action. The kinetic term arose from the Wiener measure itself, which confirms that the Gaussian character of the Wiener measure is the direct consequence of the kinetic energy on path space. Note that if $s(\tau) = 0$, the elliptic equation reduces to a diffusion equation, *i.e.* a Wick-rotated Schrödinger equation. By comparing with eq. (8), if we choose initial condition $\Psi(\xi,0) = \delta(\xi - q_i)$, the integral exactly becomes the Euclidean transition amplitude of the system from q_i to q between 0 and t. Euclidean path integrals are thus rigorous solutions of the diffusion equations, so we *expect* that, upon Wick rotation, Feynman path integrals are solutions of the Schrödinger equation.

Extending the Wiener measure. Importantly, the Feynman-Kac formula only applies to the non-relativistic Euclidean worldline theory. We would like to extend it to *complex* measures on the one hand, and to relativistic *quantum field theories* on the other hand.

Fortunately, complex measures yielding path integral solutions of Schrödinger's equation can be constructed without too much ado [41, 42] (see also ref. [43] for a more pedagogical approach), so that we do not systematically depend on Wick rotation to give a rigorous definition of non-relativistic Feynman path integrals.

The case of quantum field theories, where paths become functions defined over spacetime for example, is more intricate. It remains possible to obtain the measure associated with a free Euclidean field theory (whose action is quadratic in the fields) by constructing a Gaussian measure over a Fréchet space representing the space of fields [44], but the case of interacting field theories remains an open problem, except in some very restrictive cases such as low-dimensional theories [45], or in the perturbative limit where the interaction can be treated as a small deviation from the Gaussian measure. This gives rise to Feynman diagram expansions.

The direct construction of Feynman integrals for free and quasi-free field theories without resorting to Wick rotation has been undertaken in refs. [46, 47].

A rather panoptic account of the history of attempts to the rigorous construction of both Euclidean and Feynman integrals is given by Albeverio in the introduction of his excellent book [16].

Regularity of contributing histories. The Wiener measure is regarded as a probability measure against a class of diffusion processes (stochastic solutions of a diffusion equation) called Itō processes, of which Brownian motions (also called Wiener processes in this context) are the special case that interests us. As such, path integrals satisfy a number of identities of stochastic calculus, such as the Chapman-Kolmogorov-Smoluchowski equation for transition amplitudes

$$\int_{q(0)=q_i}^{q(t)=q_f} \mathcal{D}q\ \mathrm{e}^{\mathrm{i}S/\hbar} = \int \mathcal{D}q' \int_{q_1(0)=q_i}^{q_1(t')=q'} \mathcal{D}q_1\ \mathrm{e}^{\mathrm{i}S[q_1]/\hbar} \int_{q_2(t')=q'}^{q_2(t)=q_f} \mathcal{D}q_2\ \mathrm{e}^{\mathrm{i}S[q_2]/\hbar} \tag{60}$$

which comes from the Wiener process having the Markov property, and expresses the unitarity of time evolution. This is exactly the sewing law (12) applied to non-relativistic worldline theories.

Itō processes also satisfy Itō's lemma, which implies that *continuous paths are almost surely nowhere differentiable* [48]. This property remains true for the non-relativistic complex path integral measure [30], where it implies the canonical commutation relations, as discussed in subsection 1.4 and proved in ref. [29]. The situation worsens for higher-dimensional parameter spaces $\mathcal{M}$, *e.g.* for strings [49] or quantum fields [50]. As we said before, the free theory measure is typically defined over a Fréchet space, which is thought of as a space of distributions. In particular, it does contain locally integrable functions [51], but these form a measure zero subset [52] and thus do not contribute to the path integral. Therefore, the contributing histories are not just nowhere differentiable: they are not even functions, but singular distributions! Defining the interaction action so that it makes sense on such distributions becomes quite the challenge, and this is the main reason for the difficulties of obtaining a path integral measure for interacting QFTs. Trying to overcome these difficulties is an ongoing field of research known as *constructive quantum field theory* [53, 54, 55].

There has been some effort to obtain integrals over differentiable paths, which deviate from traditional quantum mechanics at small time scales, as should be expected [56].

Beyond the physics of thermalised systems, the idea of studying Euclidean theories comes from the common practice of *analytically continuing* functions of imaginary time to real time. The idea is simple: hopefully, Euclidean quantities are analytic functions of imaginary time, and their power series expansion has sufficient radius of convergence in the complex plane so that the corresponding correct quantum quantity can be obtained just by formally replacing imaginary time by real time [57]. The natural question is then: can we always safely perform such analytic continuations?

2.2 The Osterwalder-Schrader theorem

It is not clear in general whether analytic continuations to real time of Euclidean path integrals do exist, and so whether Wick rotation should be an allowed procedure for computing quantum quantities. The Osterwalder-Schrader theorem [58, 59] precisely provides the necessary and sufficient conditions for a set of Euclidean correlation functions to be analytically continued to Minkowski space, where the quantum field theory is defined. It really is the rigorous mathematical justification of the heuristic replacement

$$\mathrm{e}^{\mathrm{i}S[\phi]/\hbar} \quad \mapsto \quad \mathrm{e}^{-S_E[\phi]/\hbar} \tag{61}$$

Before stating the theorem, let us simply recall precisely that the observables of interest for a quantum field theory in Minkowski language are the vacuum expectation values of field operator products, which take the general form $\langle 0|\,\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\,|0\rangle$, with $|0\rangle$ the vacuum state and $\hat{\phi}$ field operators [60]. These are *Wightman distributions*, understood as distributions in the sense of Schwartz [51], defined on the n-fold product of Minkowski space. Knowing all Wightman distributions is equivalent to knowing the full quantum theory since scattering amplitudes and energy spectra, among all quantities of interest, can be derived from them. After Wick rotation, the corresponding correlation functions are called *Schwinger functions*. They are formally related to the Wightman distributions by analytic continuation of the time variables.

The question addressed by the Osterwalder-Schrader theorem is the following: given a set of Schwinger functions $\{S_n\}$, $n \geq 0$, computed in Euclidean framework, under what conditions does there exist a QFT whose Wightman distributions are obtained by analytic continuation of the S_n? To allow such a reconstruction, the functions must satisfy some axioms:

- Each S_n function must be a tempered distribution, meaning that it does not grow faster than a polynomial at infinity. This ensures that Fourier transforms exist and can be used to define momentum-space quantities;
- They must be invariant under the full connected Euclidean group (consisting in rotations, translations and reflections) so that the resulting Minkowski theory is Poincaré invariant. Upon analytic continuation, some rotations in the Euclidean plane become Lorentz boosts and invariance under Euclidean translations becomes invariance under spacetime translations;
- At large Euclidean separation, correlation functions must factorize, so we should be able to write $S_{n+m} = S_n S_m$. We indeed expect spatially separated experiments to be asymptotically independent, and this corresponds in Minkowski framework to uniqueness of the vacuum;
- The last condition must be enforced in order for the resulting Minkowski theory to have a Hilbert space with positive definite inner product. The Schwinger functions must be reflection positive, meaning that if we define the Euclidean time reflection $\theta(x_E^0, x^i) = (-x_E^0, x^i)$ and consider the smeared functional

$$A(f) = \sum_n \int \mathrm{d}x_1 \cdots \mathrm{d}x_n \, S_n(x_1, \dots, x_n) f_n(x_1, \dots, x_n) \tag{62}$$

  for test functions f defined for $x^0 > 0$, then $\langle A(\theta f)^*, A(f)\rangle \geq 0$. The functions should also be symmetric under permutation of identical fields for bosonic fields, and antisymmetric for fermion fields. In this definition, we have used the above property that the correlation functions must factorize.

If a set of Schwinger functions satisfy these axioms, the theorem guarantees they are in bijection with a Wightman QFT. Said more concretely, there exists a Hilbert space of physical states with a positive definite inner product, a unitary representation of the Poincaré group acting on it, a unique vacuum state that is invariant under action of the Poincaré group, and field operators satisfying locality and spectral conditions.

In general, the reconstruction procedure starts with building the space of functionals of Euclidean fields that are supported at positive times. Using reflection positivity, we endow it with a sesquilinear form defined on these functionals and quotient out its null space. The resulting space is then completed in the induced norm and the Hamiltonian is identified with the generator of the Euclidean time translations. This way, one can construct the physical Hilbert space. Note that when the Euclidean path integral exists as a genuine probability measure, the axioms are naturally verified, and so Wick rotation is perfectly controlled. In practice, however, the axioms can be very difficult to verify and some four-dimensional interaction theories still have not been shown to close them [61].

We now know under what conditions we can make quantum sense of imaginary time computations, but still we assumed that we can construct functional integrals to start with. A completely different and more abstract approach is to abandon the construction of the measure and instead focus on the properties that any sound path integral should satisfy.

2.3 Functorial quantum field theories

Some of the properties described in section 1, like the sewing law for example, motivated an abstract construction of path integrals in the language of category theory [62, 63]. A very nice introduction to the subject can be found in ref. [64]. In this framework, the path integral is not interpreted as a measure over fields anymore, but as a functorial assignment.

Consider as a natural starting point the path integral over a spacetime region. Let $U \subset \mathcal{M}$ be a d-dimensional spacetime domain with boundary ∂U. For a field configuration φ on ∂U, the path integral assigns a complex number

$$Z(U)(\varphi) = \Psi(\varphi) = \int_{\phi|_{\partial U}=\varphi} \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S/\hbar} \tag{63}$$

where ϕ runs over all bulk configurations consistent with φ. Physically, $\Psi(\varphi)$ encodes the probability amplitude of the boundary configuration, Z therefore maps a wavefunctional to each spacetime domain U.

At this point, a two-level structure emerges from the discussion. First, we see that Z maps each $(d-1)$-dimensional worldslice $\mathcal{S} \subset \mathcal{M}$ to the space of wavefunctionals on $\mathcal{S}$, which is the Hilbert space $\mathcal{H}_{\mathcal{S}}$ of quantum states at the time of $\mathcal{S}$, and then maps each spacetime region U such that $\partial U = \mathcal{S}$ to a wavefunctional in this Hilbert space:

$$Z \;:\; \mathcal{S} \mapsto \mathcal{H}_{\mathcal{S}}, \qquad\qquad Z \;:\; U \mapsto (\Psi \in \mathcal{H}_{\partial U}) \tag{64}$$

As a vector of $\mathcal{H}_{\partial U}$, $Z(U)$ can also be seen as a linear map from $\mathbb{C}$ to $\mathcal{H}_{\partial U}$. Generalising this, if ∂U is not just a worldslice but a disjoint union[2] $\mathcal{S}_1^* \sqcup \mathcal{S}_2$, respectively the past and future boundaries of U, then we impose $Z(U)$ to be a linear map from $\mathcal{H}_{\mathcal{S}_1}$ to $\mathcal{H}_{\mathcal{S}_2}$

$$Z(\mathcal{S}_1) \xrightarrow{Z(U)} Z(\mathcal{S}_2) \tag{65}$$

or, equivalently

$$\mathbb{C} \xrightarrow{Z(U)} Z(\mathcal{S}_1)^* \otimes Z(\mathcal{S}_2) \tag{66}$$

by simple tensor algebra. Then $Z(U)$ is defined as a vector of $Z(\mathcal{S}_1)^* \otimes Z(\mathcal{S}_2)$. Imposing that eq. (64) remains true, we find by identification

$$\mathcal{H}_{\mathcal{S}_1^* \sqcup \mathcal{S}_2} = Z(\mathcal{S}_1^* \sqcup \mathcal{S}_2) = Z(\mathcal{S}_1)^* \otimes Z(\mathcal{S}_2) = \mathcal{H}_{\mathcal{S}_1} \otimes \mathcal{H}_{\mathcal{S}_2} \tag{67}$$

The Hilbert space of composite boundaries is just the tensor product of the individual Hilbert spaces.

This extends in a straightforward way to d-dimensional domains

$$Z(U_1 \sqcup U_2) = Z(U_1) \otimes Z(U_2) \tag{68}$$

since, by locality of the classical action, if ϕ runs over two disjoint domains U_1, U_2, then the action can be written $S[\phi] = S_1[\phi_1] + S_2[\phi_2]$, where ϕ_1, ϕ_2 are respectively defined over U_1, U_2. Then

$$\int \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} = \int \mathcal{D}\phi_1 \; \mathcal{D}\phi_2 \; \mathrm{e}^{\mathrm{i}(S_1[\phi_1]+S_2[\phi_2])/\hbar} = \int \mathcal{D}\phi_1 \; \mathrm{e}^{\mathrm{i}S_1[\phi_1]/\hbar} \times \int \mathcal{D}\phi_2 \; \mathrm{e}^{\mathrm{i}S_2[\phi_2]/\hbar} \tag{69}$$

which is exactly eq. (68) with the translation of eq. (63).

[2] The asterisk indicates reverse orientation for manifolds, and dual space for vector spaces.

We obtain a very nice picture in which Z realises a systematic correspondence between two worlds: first, between $(d-1)$-dimensional submanifolds (representing boundaries) and Hilbert spaces, then between d-dimensional manifolds connecting these boundaries and linear maps connecting these Hilbert spaces, between disjoint unions and tensor products, and finally between orientation reversal and duality of Hilbert spaces. Note that we usually choose the boundaries to be compact because we can always one-point-compactify the physical space since we do not care about what happens at infinity. We say that closed $(d-1)$-manifolds are the objects of a category whose arrows are d-dimensional cobordisms between them. We say similarly that Hilbert spaces are the objects of a category whose arrows are linear maps. Arrows can be composed: cobordisms with a common boundary can be glued to obtain another cobordism, while linear maps that respectively end in and come from a common Hilbert space can be composed to get another linear morphism. Both categories are further endowed with the structure of a so-called monoidal product: disjoint union and tensor product.

We have to introduce one last ingredient: Z preserves the composition of arrows. Indeed, if U_1, U_2 are contiguous spacetime domains such that $\partial U_1 = \mathcal{S}_1^* \sqcup \mathcal{S}'$ and $\partial U_2 = \mathcal{S}'^* \sqcup \mathcal{S}_2$, then

$$Z(U_2 \cup U_1) = Z(U_2) \circ Z(U_1) \tag{70}$$

which becomes exactly the sewing law of eq. (12) when we apply eq. (63). With this additional ingredient, Z is called a *monoidal functor* between the aforementioned categories. Different functors then correspond to different theories.

This construction has the great merit of dodging the difficult problem of the functional measure. Its main drawback is that it does not construct the path integral of a specific theory, but rather gives a general form for all possible path integrals. This has allowed the classification of, *e.g.*, topological quantum field theories (TQFT) [65, 66, 67].

Each of the mathematical approaches outlined in this section has its strengths, but also its open problems that prevent it from generalising to any physical path integral. Despite these limitations, or because of them, much information can be extracted from path integrals by using formal rules which are not fully mathematically grounded. Such formal rules, heuristic arguments and methods have proved extremely productive, and we focus now on these physically rich developments.

3 Rules and methods of the path integral calculus

In this section, we enumerate various rules used in the literature to formally handle path integrals. We begin in subsection 3.1 by studying the long-time limit, in which path integral expectation values become stationary state expectation values. Subsection 3.2 is dedicated to the time-slicing procedure, which is usually used to define non-relativistic path integrals, and introduces its lattice-based generalisation to field theories. Subsection 3.3 deals with the quantisation of constrained systems through the simple insertion of a Dirac delta pseudo-functional. In subsection 3.4, a rough outline of the derivation of perturbative and semi-classical expansions of quantum-mechanical quantities is used to discuss the structure of the classical-to-quantum transition. Subsection 3.5 shows how quantum information in path integrals can be redistributed by changes of variables of very diverse kinds, and eventually, perturbative expansions, semi-classical approximations and changes of variables are all shown to be important ingredients of the more general setting of effective theories, which are introduced in subsection 3.6.

As one might see from the description of this section, our aim is to go beyond the technical aspects of typical path integral methods and underline what each of them reveals about the nature of quantum theories. By describing them in the common formalism of path integrals and by showing how they naturally lead to the general framework of effective theories, we hope to make clear that these methods are not independent tools but rather rely on physical approximations that can be efficiently controlled and compared if correctly understood.

3.1 Long-time limit

We may wonder by now which configuration histories in $\mathcal{C}$ indeed contribute to arbitrary path integrals. We already said in sections 1.4, 2.1 that nowhere differentiable histories near the classical ones contribute the most, but we should precise what "near" means in this context. Consider for instance two histories ϕ_c, ϕ' with initial energy E: one is classical and thus its energy remains conserved in time t, the other is non-classical and its energy undergoes a change of the order of ΔE. Then the action changes by

$$\Delta S = \int_0^t \mathrm{d}\tau\, E(\phi', \dot{\phi}') - \int_0^t \mathrm{d}\tau\, E(\phi_c, \dot{\phi}_c) \sim t\Delta E \tag{71}$$

Then, if $t\Delta E \gg \hbar$, the action difference becomes $\Delta S \gg \hbar$, the complex exponential oscillates very rapidly when integrating between ϕ_c and ϕ' and the contribution of ϕ' vanishes by destructive interference. The histories that contribute are precisely those for which $t\Delta E \lesssim \hbar$, which means there is a fundamental indeterminacy of the order ΔE on the energy of the quantum system obtained by summing over all of these histories:

$$t\Delta E \gtrsim \hbar \tag{72}$$

This is a heuristic argument that shows how Heisenberg's indeterminacy principle arises in the path integral formalism.

This means that we can restrict the space of configuration histories $\mathcal{C}$, without loss of information, to a region containing the (possibly multiple) classical histories, and of energy range $\Delta E \sim \hbar/t$. Now, if we take time integration from $-\infty$ to $+\infty$, we enforce $\Delta E \to 0$, meaning that only histories with exactly constant energy survive. For smooth histories, this criterion would select only the classical solution, but we already know that contributing histories are not smooth at all, and thus it instead selects nowhere differentiable histories that are infinitely close to the classical path.

Among all of the constant-energy histories, those with the lowest energy will dominate. Histories of higher constant energies are further from the classical history, so their contributions will be exponentially suppressed compared to the ground state. This is again a heuristic argument, but it leads us to postulate:

Postulate (Long-time principle)**.** *Let $|0\rangle$ be the ground state of the system of action $S = \int_0^t L\,\mathrm{d}\tau$ (where L is the classical Lagrangian). Then, the ground state expectation value of an observable O is the long-time limit of its expectation value against the set $\mathcal{C}$ of configuration histories*

$$\langle 0|\hat{O}|0\rangle = \lim_{t\to\infty} \langle O\rangle_{\mathcal{C}} = \int_{\mathcal{C}} \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}\int_{-\infty}^{+\infty} L\mathrm{d}\tau} O[\phi] \tag{73}$$

This postulate can actually be proved in the Euclidean framework where, as a reminder, a Wick rotation $t \mapsto -\mathrm{i}\hbar\beta$ is performed as described by eq. (50). Then:

$$\int \mathcal{D}\phi\, \mathrm{e}^{-S_E/\hbar} O[\phi] = \frac{1}{Z_E} \sum_j \langle \Psi_j|\mathrm{e}^{-\beta\hat{H}}\hat{O}|\Psi_j\rangle = \sum_j \mathrm{e}^{-\beta E_j} \langle \Psi_j|\hat{O}|\Psi_j\rangle \tag{74}$$

which is just the spectral decomposition of the Euclidean transition amplitude, obtained by inserting resolutions of the identity in the energy eigenbasis. E_j, Ψ_j are then respectively the eigenvalues and eigenstates of the Hamiltonian $\hat{H}$. Taking the low-temperature limit $\beta \to \infty$ which, by Wick rotation, is equivalent to the long-time limit, we obtain that all terms in the sum become very small compared to the E_0 (ground state) term. In fact, this term vanishes too, except if we choose $E_0 = 0$, which we can do without loss of generality since energy is defined up to a constant. Thus:

$$\lim_{\beta\to\infty} \int \mathcal{D}\phi\, \mathrm{e}^{-S_E/\hbar} O[\phi] = \langle 0|\hat{O}|0\rangle \tag{75}$$

Whenever Euclidean expectation values can be analytically continued to complex expectation values (when the Osterwalder-Schrader theorem holds, for instance), this proves the long-time principle.

In particular, the partition function $\int \mathcal{D}\phi\, \exp(\mathrm{i}S/\hbar)$ is just the expectation value of the constant functional equal to 1. In the long-time limit, it becomes the ground state expectation value $\langle 0|0\rangle$ up to an unimportant phase factor. This makes sense since the partition function is used to normalise path integral expectation values, just as $\langle 0|0\rangle$ normalises operator expectation values.

The long-time principle can also be derived through Feynman's $\mathrm{i}\varepsilon$ prescription: by setting the ground state energy to zero and multiplying Hamilton's function by $(1-\mathrm{i}\varepsilon)$, where ε is an arbitrarily small real number, the real-time exponentials in the spectral decomposition of the path integral are suppressed at large real times (while they would be purely oscillatory without the $\mathrm{i}\varepsilon$ term), except the term corresponding to the ground state since its energy is set to zero [68]. Although this procedure has been argued to be more general than the Euclidean time procedure [69], it remains quite mathematically involved and generates an unphysical non-hermiticity that can be difficult to address [70, 71]. Most interestingly, slightly adapting the $\mathrm{i}\varepsilon$ prescription yields long-time path integrals for *excited states* expectation values [72]. We will mainly concentrate on ground states in the following and thus do not develop this point, but the reader should keep in mind that all formal results in the long-time limit, as long as we do not assume a particular form of the action (so as not to fix a precise $\mathrm{i}\varepsilon$ prescription), are valid for excited states as well as for the ground state.

3.2 Lattice-slicing approximation

As we underlined through this paper, the main problem of formal path integrals is that they are rarely computable. This section and section 3.4 are dedicated to two complementary methods that are widely used to obtain computable approximations to formal path integrals: *lattice-slicing* is treated in this section, and *perturbation expansions* in section 3.4.

Time-slicing of worldline theories. Let us start with a non-relativistic worldline theory, for which discretisation is in fact Feynman's original insight on path integrals [2]. The procedure is called *time-slicing* and the main idea is to discretise the continuous evolution of a system into infinitesimal time steps. Consider a non-relativistic system of particles transitioning between two states $|q_i\rangle$ and $|q_f\rangle$ over a total time t. Dividing the interval into n steps of duration $\varepsilon = t/n$, the evolution operator can be decomposed into a product of short-time transition amplitudes

$$\mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar} = \lim_{n\to\infty}\left(\mathrm{e}^{-\mathrm{i}\varepsilon\hat{H}/\hbar}\right)^n \tag{76}$$

By inserting a resolution of the identity in the position basis between each intermediary time step, the transition amplitude can be expressed as a multi-dimensional integral

$$\int \mathcal{D}q(t)\ \mathrm{e}^{\mathrm{i}S/\hbar} = \langle q_f|\mathrm{e}^{-\mathrm{i}t\hat{H}/\hbar}|q_i\rangle = \lim_{n\to\infty}\int \mathrm{d}q_1\ldots\mathrm{d}q_{n-1}\prod_{k=1}^{n}\langle q_k|\mathrm{e}^{-\mathrm{i}\varepsilon\hat{H}/\hbar}|q_{k-1}\rangle \tag{77}$$

with $q_0 = q_i$ and $q_n = q_f$. The complicated infinite-dimensional functional integral is replaced by a product of well-defined standard integrals. Many analytical proofs, as well as numerical computations, are performed using this method (Kleinert's excellent book entirely relies on time-slicing arguments [6], you can notably find there a simple proof of eq. (11) for the free transition amplitude). Writing the Hamiltonian for a particle of mass m with momentum and potential $\hat{H} = \hat{p}^2/2m + \hat{V}$, each short-time transition amplitude can be further computed by inserting the identity in momentum basis and evaluating the Gaussian integrals over momenta. The fictitious functional measure then admits a closed form [73]

$$\mathcal{D}q(t) = \lim_{n\to\infty}\mathrm{e}^{-\mathrm{i}\pi n/4}\left(\frac{m}{h\varepsilon}\right)^{n/2}\prod_{k=1}^{n-1}\mathrm{d}q_k \tag{78}$$

where integrals are replaced by sums and derivatives by finite differences. Since the Lebesgue measure $\mathcal{D}q$ does not exist according to the discussion of subsection 2.1, this expression is purely formal and the limit only makes sense once inserted in the path integral

$$\int \mathcal{D}q(t)\ \mathrm{e}^{\mathrm{i}S/\hbar} = \lim_{n\to\infty}\mathrm{e}^{-\mathrm{i}\pi n/4}\left(\frac{m}{h\varepsilon}\right)^{n/2}\prod_{k=1}^{n-1}\int \mathrm{d}q_k\exp\left(\frac{\mathrm{i}\varepsilon}{\hbar}\left[\frac{m}{2}\left(\frac{q_k-q_{k-1}}{\varepsilon}\right)^2 - V(q_k,k\varepsilon)\right]\right) \tag{79}$$

This limit is proved to exist in a large variety of cases (see many examples in ref. [6]). Even when it does not exist, it may provide interesting insight on the theory.

The time-sliced action is chosen so that it converges to the classical action. It may be, however, that several time-sliced actions converge to the same classical one, but that the corresponding time-sliced path integrals do not converge to the same value. The problem arises when the action contains products of several functionals evaluated at equal time, typically terms in $q(t)\dot{q}(t)$. As we showed in subsection 1.4, we should avoid trying to insert such products in a path integral since it leads to time-ordering ambiguities. Having such products in the *action* was however not a problem in the formal path integral since the action is not inserted as an observable functional. The term $q\dot{q}$ can be time-sliced in a *forward*, *backward* or *symmetric* way:

$$q(t)\frac{q(t+\varepsilon)-q(t)}{\varepsilon},\quad q(t+\varepsilon)\frac{q(t+\varepsilon)-q(t)}{\varepsilon},\quad q(t)\frac{q(t+\varepsilon)-q(t-\varepsilon)}{2\varepsilon} \tag{80}$$

None of these choices are equivalent. The calculus of stochastic integrals with one of the two first is known as *Itō calculus*, wherein Itō's lemma implies that the difference of the two first expressions is non-zero (which is why the stochastic process is nowhere differentiable). The third choice leads to *Stratonovich calculus*, which is usually the choice made by physicists since many practical formulae such as the chain rule hold there, but break in Itō calculus [48]. There is no general prescription, however, that tells us which of these choices will yield the correct, physical path integral.

Finally, note that after Wick rotation $\varepsilon \to -i\delta\tau$, each short-time transition amplitude becomes a real, positive *transfer matrix* $\langle q_k|\mathrm{e}^{-\hat{H}\delta\tau/\hbar}|q_{k-1}\rangle$, which is the statistical physics counterpart to the quantum amplitude.

Time-slicing can in principle be applied even to path integrals with higher-dimensional parameter spaces $\mathcal{M}$. In this case, it consists in decomposing $\mathcal{M}$ in discrete worldslices rather than just points. But this only displaces the problem to one lower dimension, since the product of integrals in eq. (79) becomes a product of path integrals over the *boundary field theory*.

Lattice field theories. A better course of action to generalise time-slicing to arbitrary dimensions is then *lattice field theory* [74, 75]. In this framework, $\mathcal{M}$ is the Lorentzian or Euclidean spacetime $\mathbb{R}^d$, discretised to a hypercubic lattice

$$\Lambda = \left\{ x^\mu = an^\mu \middle| \, n^\mu \in \mathbb{Z}^d, \mu = 0, 1, \ldots, d-1 \right\} \tag{81}$$

with $a > 0$ the lattice spacing. The usual fields $\phi(x)$ are then defined on the lattice sites and they become a finite, or infinite but countable, set of degrees of freedom $\phi_n = \phi(an)$. In practice, one typically works on a finite lattice (if the lattice is of extent N in each direction, we get N^d sites) and impose periodic boundary conditions to ensure translational symmetry.

Let us take again the simplest example of all, the case of the real scalar field in d dimensions, for which the action $S[\phi] = \int \mathrm{d}^d x \left[\frac{1}{2}(\partial_\mu \phi)^2 + V(\phi)\right]$ translates in lattice field theory language to

$$S_{\text{lattice}}[\phi] = a^d \sum_{n \in \Lambda} \left[\frac{1}{2} \sum_{\mu=0}^{d-1} \left(\frac{\phi_{n+\hat{\mu}} - \phi_n}{a} \right)^2 + V(\phi_n) \right] \tag{82}$$

where $\hat{\mu}$ denotes the unit vector in the μ-direction on the lattice. The path integral is then well-defined because it is just an ordinary integral over field values at each lattice site:

$$Z = \int \mathcal{D}\phi \, \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} \quad \to \quad Z_{\text{lattice}} = \int \prod_{n \in \Lambda} \mathrm{d}\phi_n \, \mathrm{e}^{\mathrm{i}S_{\text{lattice}}[\phi]/\hbar} \tag{83}$$

In practice, one typically performs a Wick rotation to Euclidean time to be able to use Monte-Carlo methods. The Euclidean lattice partition function is then formally equivalent to the partition function of a classical statistical mechanics system in d dimensions. Whenever the Osterwalder-Schrader theorem applies, Euclidean Monte-Carlo computations can be sent back to Minkowski space to be interpreted as transition amplitudes.

This naive discretisation procedure cannot be extended too fast. For example, imposing the substitution $A_\mu(x) \to A_\mu(an)$ fails for gauge fields because local gauge invariance must be preserved on the lattice. Wilson proposed to represent the gauge field on the lattice with the *link variable*

$$U_\mu(n) = \mathrm{e}^{\mathrm{i}aA_\mu(n)} \in G \tag{84}$$

which does *not* live on the lattice sites but on the oriented links connecting neighbouring sites n and $n + \hat{\mu}$ [76]. This construction also ensures that the objects transform correctly under gauge transformations. The natural elementary gauge-invariant observable is called the *plaquette*

$$P_{\mu\nu}(n) = U_\mu(n) U_\nu(n + \hat{\mu}) U_\mu^\dagger(n + \hat{\nu}) U_\nu^\dagger(n) \tag{85}$$

i.e. the product of four link variables around an elementary square of the lattice (as illustrated in fig. 2). In the $a \to 0$ limit, it becomes

$$P_{\mu\nu}(n) \approx \mathrm{e}^{\mathrm{i}a^2 F_{\mu\nu}(n) + O(a^3)} \tag{86}$$

revealing that the plaquette really encodes the *field strength* of the field theory [77]. As G may represent an Abelian or non-Abelian group, the field strength may capture very different effects (in the non-Abelian case, $F_{\mu\nu}$ is defined as $F^a_{\mu\nu} t^a$ with t^a the group generators). The Wilson action for these plaquettes

$$S_{\text{Wilson}}[U] \propto \sum_{n \in \Lambda} \sum_{\mu < \nu} \mathrm{Re}\{1 - P_{\mu\nu}(n)\} \tag{87}$$

correctly converges to the continuous Yang-Mills action. The measure $\mathrm{d}U$ that appears in the resulting integrals is the well-known Haar measure over the gauge group G.

The discretisation of the Dirac action for fermionic fields presents other subtleties. Derivatives must be replaced by the symmetric difference (80) in order to preserve chiral symmetry [78], and the lattice action then reads

$$S_{\text{naive}}[\psi, \bar{\psi}] = a^d \sum_{n \in \Lambda} \bar{\psi}_n \left[\sum_{\mu=0}^{d-1} \hbar c \gamma^\mu \frac{\psi_{n+\hat{\mu}} - \psi_{n-\hat{\mu}}}{2a} + mc^2 \psi_n \right] \tag{88}$$

However, the symmetric difference not only vanishes for constant fields (just like a continuous derivative would), but also for example for any field that alternates sign on neighbouring sites! These rapidly oscillating modes live on the edge of the Brillouin zone and are invisible to the lattice Dirac operator, so they behave exactly as massless fields would and are indistinguishable from the physical fermions from the point of view of the action. Discretisation introduces additional artificial symmetries that generate, in d dimensions, $2^d - 1$ such modes in addition to the physical one, for a total of 2^d degenerate particles. This is the *fermion doubling problem*. Even worse, according to the Nielsen-Ninomiya theorem, it is *impossible* to have locality, chiral symmetry, and the correct physical theory in the continuum limit without fermion doubling on a local, translation-invariant lattice [79, 80]. Any solution to this problem must sacrifice one of these conditions [76].

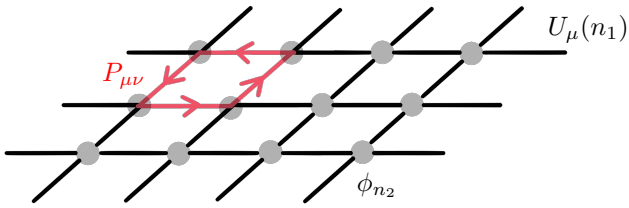

Figure 2: Illustration of the lattice construction and the field correspondance: scalar and fermion fields live on the lattice sites while the gauge fields live on the links between them. A plaquette (in red) corresponds to an oriented cell.

Continuum limit. The exact, continuum physics is recovered in the $a \to 0$ limit, but at the same time we must take $N \to \infty$ so that the volume of the system Na remains fixed. However, the lattice action is defined by a set of parameters (masses, couplings,...) that all depend on a. For the limit to make sense, these must be tuned so that physical quantities remain finite and converge to well-defined values as $a \to 0$. We can get some insights from statistical mechanics. In a statistical system, the correlation length ξ defines the typical distance over which fluctuations remain correlated. Away from any phase transition, ξ is finite, so correlations decay exponentially with spacing. a thus sets the only length scale of the system and any physical prediction implicitly retains this information. The continuum limit requires $\xi/a \to \infty$ so that the physics at large distance becomes completely insensitive to the lattice structure, which is precisely the behaviour of a *second order phase transition*! The continuum limit must thus be taken by tuning the lattice parameters towards the values corresponding to a second order critical point. We will see that a similar discussion arises in subsection 3.6 as lattice theories can be seen as low-energy effective theory analogues. Not every theory possesses such a critical point, for example, the four-dimensional Yang-Mills theory lacks a proof of the existence and uniqueness of its continuum limit [61].

3.3 Constrained quantisation and functional determinants

Path integrals admit a natural implementation of quantisation of constrained systems, a problem that is not so straightforward in canonical quantisation [81]. Before delving in the subject, we will extend our multi-indices from section 1.4 to a convenient notation for functional calculus.

Notation (DeWitt)**.** Let Λ be a multi-index running over all possible indices in $\mathcal{N}$ (including, possibly, complex and assimilated conjugations). Then a *DeWitt index* $i = (\Lambda, x)$ contains both the multi-index of $\mathcal{N}$ and the parameters in $\mathcal{M}$. Differentiation becomes

$$F_{,i} = \frac{\delta F}{\delta \phi^i} = \frac{\delta F}{\delta \phi^\Lambda(x)} \tag{89}$$

and index contraction contains both summation and integration

$$F_i G^i = \sum_\Lambda \int_{\mathcal{M}} \mathrm{d}x\, F_\Lambda(x) G^\Lambda(x) \tag{90}$$

To understand the rationale behind constrained quantisation, let us imagine that the full history space $\mathcal{C}$ is homeomorphic to a Cartesian product $\mathcal{C}' \times \mathcal{F}$, where $\mathcal{C}'$ is the subspace of constrained configuration histories, and $\mathcal{F}$ describes the supplementary degrees of freedom of the unconstrained system, with integration variable $U \in \mathcal{F}$. The path integral of the constrained system should be

taken over $\mathcal{C}'$, however this space is often complicated to parametrise so we would rather perform the integration over $\mathcal{C}$, and then project onto $\mathcal{C}'$ using a delta functional to enforce the constraint. For each constrained configuration ϕ', we choose a unique representative $U_0[\phi']$ for the redundant variable U. This choice is completely arbitrary, and carries a DeWitt index since it must fix U^i for each value of i (at east for each point x in $\mathcal{M}$, and possibly for other indices of $\mathcal{N}$). Then:

$$\int_{\mathcal{C}'} \mathcal{D}\phi' \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} = \int_{\mathcal{C}'} \mathcal{D}\phi' \int_{\mathcal{F}} \mathcal{D}U \ \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} \ \delta[U^i - U_0^i[\phi']] \tag{91}$$

However, we do not know in general how to write $U_0[\phi']$ explicitly, so instead, we encode the constraint in a constraint functional $C^i[\phi', U]$ that vanishes whenever $U^i = U_0^i[\phi']$. According to the functional equivalent of a standard formula for delta functions, we have

$$\delta[C^i[\phi', U]] = \delta[U^i - U_0^i[\phi']] \times \left|\det\left(\frac{\delta C^j}{\delta U^k}\right)\right|^{-1} \tag{92}$$

This expression involves a highly non-trivial **functional determinant**, that is formally defined for an arbitrary operator $K^i{}_j$ by [32]

$$\det\big(K^i{}_j\big) = \left[\int \mathcal{D}c \ \mathcal{D}c' \ \mathrm{e}^{\mathrm{i}c'_i K^i{}_j c^j/\hbar}\right]^{-1} \quad \text{if } c, c' \text{ are Grassmann-even (commuting variables)} \tag{93a}$$

$$\det\big(K^i{}_j\big) = \left[\int \mathcal{D}c \ \mathcal{D}c' \ \mathrm{e}^{\mathrm{i}c'_i K^i{}_j c^j/\hbar}\right]^{+1} \quad \text{if } c, c' \text{ are Grassmann-odd (anticommuting variables)} \tag{93b}$$

These definitions are formal since a rigorous extension of the notion of determinant to infinite-dimensional spaces implies regularisation techniques which do not want to enter here [82]. By contrast, the expressions we give here are typically divergent, and only make sense once they are divided by appropriate normalisation factors. For most of the formal calculus, manipulating only these formal expressions is sufficient. Besides, one may find in the literature an additional minus sign or no imaginary unit in the exponentials of eq. (93), this is an indifferent choice of convention since it only contributes a global phase to the resulting determinant, which may be absorbed in the anyway divergent normalisation factor.

Taking eq. (92) into account and injecting that $\mathcal{C}' \times \mathcal{F} \cong \mathcal{C}$, the constrained path integral becomes

$$\int_{\mathcal{C}'} \mathcal{D}\phi' \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} = \int_{\mathcal{C}} \mathcal{D}\phi \ \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} \ \delta[C^i[\phi]] \left|\det\left(\frac{\delta C^j}{\delta U^k}\right)\right| \tag{94}$$

Moreover, considering the Fourier representation of the Dirac delta pseudo-functional

$$\delta[C[\phi]] = \int \mathcal{D}\lambda \ \mathrm{e}^{\mathrm{i}\lambda_i C^i[\phi]/\hbar} \tag{95}$$

and the definition (93b) of the functional determinant involving Grassmann-odd histories, we get the final result (in the case where the determinant is positive) [33, 19, 83]:

$$\int_{C[\phi]=0} \mathcal{D}\phi \ \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} = \int \mathcal{D}\phi \ \mathcal{D}\lambda \ \mathcal{D}c \ \mathcal{D}c' \ \exp\left(\frac{\mathrm{i}}{\hbar}\left(S[\phi] + \lambda_i C^i[\phi] + c'_j \frac{\delta C^j}{\delta U^k} c^k\right)\right) \tag{96}$$

The new variable λ, called the *Nakanishi-Lautrup field* in the context of constrained quantisation, precisely plays the rôle of a Lagrange multiplier, so that the corresponding new term $\lambda_i C^i[\phi]$ in the action enforces the constraint $C[\phi] = 0$ in the classical equations of motion. The Dirac delta functional is however not invariant upon rescaling of the constraint equation $C[\phi]$ and so the determinant $\det\big(\delta C^j/\delta U^k\big)$ (called the *Faddeev-Popov determinant*) is needed to absorb this unphysical rescaling dependence. The Grassmann-odd fields c, c' are called respectively Faddeev-Popov ghosts and antighosts. They are usually denoted $c, \bar{c}$ but we wanted to prevent the confusion with Dirac conjugation: these fields are *a priori* not spinors. In many cases, they are anticommuting scalar fields (so fermions of spin 0 in QFT!), which makes them a good example of the limitations of the spin-statistics theorem, which only applies to observable configuration variables.

This method is useful for path integrals over gauge fields. As explained in section 1.5, gauge symmetries reflect a redundancy in our description of physics, where a class of gauge equivalent fields in fact captures the same physical state (think *e.g.* of the electromagnetic field generated by a gauge potential). Naively integrating over the whole, redundant space of gauge fields A would result in irrecuperable divergences. The standard way-out is to impose a *gauge-fixing condition* $C[A] = 0$ through the method presented above [84]. We will see an interesting application of Lagrange multipliers for gauge fields in section 4.3.

3.4 Semi-classical approximation and perturbative expansion

In practice, expanding around some solvable limit can be achieved through two main approaches: the *semi-classical* expansion, where the theory is expanded in powers of $\hbar$ around classical solutions, and the *perturbative expansion*, expanded in powers of a coupling constant around a solvable free theory. While they are often treated separately, they are closely related and can be understood in the same framework.

Assume first that the action $S[\phi]$ admits a classical solution ϕ_c. By expanding the field around this saddle point $\phi = \phi_c + \sqrt{\hbar}\eta$, where η denotes a small displacement in field space, we can perform a functional Taylor expansion of the action (in DeWitt's notation)

$$S[\phi] \approx S[\phi_c] + \frac{\hbar}{2}\eta^i S_{,ij}\eta^j \tag{97}$$

and the rest of the expansion contains higher-order terms in η. At first order in $\hbar$, the integral simplifies to a Gaussian functional integral:

$$Z = \int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} \approx \mathrm{e}^{\mathrm{i}S[\phi_c]/\hbar} \int \mathcal{D}\eta\, \mathrm{e}^{\mathrm{i}\eta^i S_{,ij}\eta^j/2} \tag{98}$$

and the last factor can be evaluated exactly to yield a functional determinant.

In this regime, it is clear that the classical action determines the overall phase of the configuration while the functional determinant encodes how fluctuations spread around the classical trajectory. This can be seen as the functional analogue of the method of stationary phase. As we briefly discussed in subsection 1.1, the validity of the approximation is controlled by the ratio $S[\phi_c]/\hbar$, and if this ratio is large, the integrand oscillates quickly in configuration space which generates destructive interferences. These suppress contributions of configurations far from the saddle and so localise the integral near stationary points. Quantum corrections thus appear as a systematic expansion in powers of $\hbar$. However, when the action is comparable to $\hbar$, no single saddle dominates and the classical picture loses its meaning. It is worth noting that nothing forces the saddle to be unique, so in general, we must sum over all contributing saddle points and interferences can generate non-perturbative effects such as tunnelling.

The other situation arises when the action can be decomposed as a solvable part (*i.e.* an action whose integral we can compute) and a small perturbation:

$$S[\phi] = S_0[\phi] + S_{\mathrm{pert}}[\phi] \tag{99}$$

We can then write the path integral as

$$Z = \int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S_0[\phi]/\hbar}\, \mathrm{e}^{\mathrm{i}S_{\mathrm{pert}}[\phi]/\hbar} \tag{100}$$

and expand the perturbation term as a power series:

$$\mathrm{e}^{\mathrm{i}S_{\mathrm{pert}}[\phi]/\hbar} = \sum_{k=0}^{\infty} \frac{1}{k!}\left(\frac{\mathrm{i}}{\hbar}S_{\mathrm{pert}}\right)^k \tag{101}$$

Yet, we saw in previous discussions that the expectation value of any functional $F[\phi]$ with respect to the free theory is defined as

$$\langle F[\phi]\rangle_0 = \frac{1}{Z_0}\int \mathcal{D}\phi\, F[\phi]\mathrm{e}^{\mathrm{i}S_0/\hbar} \tag{102}$$

where Z_0 is the zeroth order partition function $\int \mathcal{D}\phi\, \exp(\mathrm{i}S_0/\hbar)$. This allows for the full partition function to be re-expressed as the expectation value of the perturbation term

$$Z = Z_0 \left\langle \mathrm{e}^{\mathrm{i}S_{\mathrm{pert}}[\phi]/\hbar}\right\rangle_0 = Z_0 \sum_{k=0}^{\infty}\frac{1}{k!}\left\langle \left(\frac{\mathrm{i}}{\hbar}S_{\mathrm{pert}}\right)^k\right\rangle_0 \tag{103}$$

and now each term further reduces to expectation values of the action computed in the solvable theory. Whenever S_{pert} is small, only the first terms in the expansion contribute to the result.

In QFT, S_0 is often chosen to be the action of the free theory, which is quadratic in the field and makes the integral exactly Gaussian, and S_{pert} contains interaction terms, typically scaled by a (small) coupling constant λ. These computations are tractable using Wick's theorem which expresses

any of the expectation values in eq. (103) as a sum over all possible pairings of the fields, so all the work is just a sequence of Gaussian computations [85]. In both cases, what is important is that we were able to separate the computation of the partition function. The celebrated formalism of *Feynman diagrams* [86, 87] appears naturally once the path integral is written as a Gaussian core (the free, solvable theory) plus interactions. The Gaussian part describes the propagation of free fluctuations because, since S_0 is quadratic, the path integral can be computed exactly and gives the probability amplitude for the fluctuation to travel between two spacetime points. This is usually called the *Feynman propagator*. In order to reconnect with textbook notations, for a real scalar field of mass m, it takes the closed form

$$\Delta_F(x-y,m) \equiv \langle\phi(x)\phi(y)\rangle_0 = \int \frac{\mathrm{d}^4p}{(2\pi\hbar)^4}\frac{\mathrm{i}}{p^2-(mc)^2+\mathrm{i}\varepsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)/\hbar} \tag{104}$$

where the $\mathrm{i}\varepsilon$ comes from the prescription used to obtain the long-time principle (see the discussion in subsection 3.1). The interaction part represents local events where several fluctuations meet and interact, and its expansion corresponds to considering processes with increasing numbers of such interactions. Diagrams then provide a visualisation of quantum processes according to the so-called *Feynman rules*:

- A *line* represents the propagation of a fluctuation between points x and y, and is weighted by the value of the propagator;
- *Vertices* correspond to points where several fluctuations interact. Each vertex reflects the form of the interaction (a vertex describing an η^4 interaction will have four legs) and carries a unit of the coupling constant.

An example of diagram expansion is shown in fig. 3. Then:

Theorem (Asymptotic expansion in Feynman diagrams)**.** *The partition function can be asymptotically expanded in terms of Feynman graphs* Γ*:*

$$Z \sim Z_0 \sum_\Gamma A(\Gamma)\ \hbar^{e(\Gamma)-v(\Gamma)}\ (-\lambda)^{v(\Gamma)} \tag{105}$$

where $A(\Gamma)$ *is the amplitude obtained by Feynman rules, that only depends on external parameters and is normalised by the number of graphs equivalent to* Γ*,* $e(\Gamma)$ *and* $v(\Gamma)$ *are respectively the number of edges and vertices of the graph, and* λ *is the coupling constant of the theory [19].*

This formula can be extended to the case of several different coupling constants. According to Euler's theorem, for connected graphs (which are not the only ones that contribute to Z, however), the difference $e(\Gamma)-v(\Gamma)$ is just the number of loops in the graph. Thus, we immediately see that the kth order in the semi-classical expansion is given by k-loop diagrams, while the kth order in the perturbative expansion is given by diagrams with k vertices.

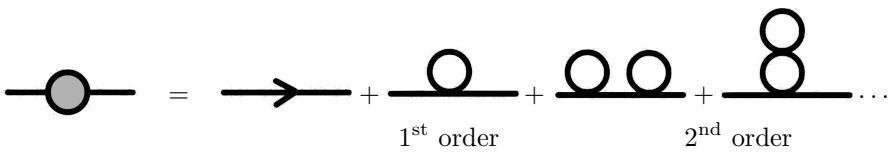

Figure 3: Diagrammatic perturbative expansion of the propagation of a scalar field (only part of the 2nd order is represented).

In both cases, the leading order is given by *tree diagrams*, defined as diagrams with no loop and minimal number of vertices, which is zeroth order in $\hbar$ and thus corresponds to the classical theory. This explains why in eq. (11) the free theory partition function was exactly obtained by evaluating the action in the classical history. On the other hand, loops are responsible for introducing quantum fluctuations around a classical solution and must be handled with care because they encode short-distance physics which usually requires renormalisation and regularisation.

Perturbative expansions are in general asymptotic rather than convergent. The factorial growth of the number of diagrams at high order implies that any finite truncation misses contributions from distinct classical saddles and effects such as tunnelling. It is clear from this and from eq. (105) that for *completely quantum* theories (like highly entangled systems for example), *both* the semi-classical and perturbation expansions fail dramatically. Perturbative path integrals, however, often have the

nice property of being well-defined, since they represent only small deformations of the well-defined Gaussian free theory.

From the point of view adopted in this article, perturbative and semi-classical approximations reveal the structure of the classical-to-quantum transition, where the phase is governed by the classical action, the quadratic term defines the free part of the theory and the higher-order terms generate combinatorial fluctuations. We do not focus here on reviewing Feynman diagram methods, so we refer the reader to the very complete and pedagogical refs. [10, 88].

3.5 Changes of variables

The path integral formalism also accommodates transformations of the integration variables, which can be used to simplify (formal) computations or study symmetries of the quantum theory. In principle, by analogy with changes of variables in standard integrals, a change of variable should be understood as a diffeomorphism between two spaces of histories

$$\begin{aligned} f \quad : \quad & \mathcal{C} \longrightarrow \mathcal{C}' \\ & \phi \longmapsto \phi' \end{aligned} \tag{106}$$

that leaves the value of a path integral unchanged. Since $\mathcal{C}$ and $\mathcal{C}'$ are by definition diffeomorphic, we will identify them in the following. f is required to be a diffeomorphism (invertible and sufficiently smooth) since we want to be able to define and evaluate the **functional Jacobian matrix**:

$$\frac{\delta\phi'(y)}{\delta\phi(x)} = \left.\frac{\delta f[\phi]}{\delta\phi(x)}\right|_y \quad \text{or more generally} \quad f^j{}_{,i} \tag{107}$$

whose determinant J is expected to measure the change in volume associated with the change of variable. Upon such a transformation, the path integral takes the form

$$\int_{\phi|_{\partial\mathcal{M}}=\varphi} \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} O[\phi] = \int_{\phi'|_{f[\partial\mathcal{M}]}=f[\varphi]} \mathcal{D}\phi' \; J[\phi'] \; \mathrm{e}^{\mathrm{i}f_*S[\phi']/\hbar} f_* O[\phi'] \tag{108}$$

where we write $f_*O[\phi'] = O[f^{-1}[\phi']]$ the pushforward of O along f to alleviate notations. For simple translations $\phi \mapsto \phi + \chi$ with χ a fixed background history, the Jacobian is trivial and the measure is left unchanged. However, for non-linear transformations, the Jacobian is generally non-trivial, and if it can be written as a complex exponential, it contributes additional terms to the action [89].

This procedure is directly used to study the behaviour of quantum theories under continuous symmetries of the classical action. Assume the classical action is invariant, up to boundary terms, under some *global* infinitesimal transformation $\phi \mapsto \phi + \delta\phi$ (*i.e.* $\delta\phi$ is independent of x). Then apply a *local* infinitesimal transformation of the fields $\phi(x) \mapsto \phi(x) + \delta\phi(x)$ to the path integral expectation value of an arbitrary functional $\langle O^i \rangle_{\mathcal{C}}$. The action is of course not invariant anymore, but its derivatives with respect to the local transformation yield the spacetime derivatives of the Noether currents $S_{,j} = -\partial_\mu I^\mu{}_j$ (see *e.g.* ref. [90], §2.2). If the fictitious measure is also invariant under this transformation, one obtains

$$\begin{aligned} \int_{\mathcal{C}} \mathcal{D}\phi \; \mathrm{e}^{\mathrm{i}S[\phi]/\hbar} O^i[\phi] &= \int_{\mathcal{C}} \mathcal{D}\phi' \; \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} \left(1 + \frac{\mathrm{i}}{\hbar}\delta S\right) \left(O^i[\phi'] + \delta O^i\right) \\ &= \int_{\mathcal{C}} \mathcal{D}\phi' \; \mathrm{e}^{\mathrm{i}S[\phi']/\hbar} O^i[\phi'] + \int_{\mathcal{C}} \mathcal{D}\phi' \; \mathrm{e}^{\mathrm{i}S/\hbar} \left(\delta O^i + \frac{\mathrm{i}}{\hbar} O^i \delta S\right) \end{aligned} \tag{109}$$

which, after cancelling the l.h.s. with the first term in the r.h.s. and expanding the functional differentials in functional derivatives, yields the (non-anomalous) *Ward-Takahashi identities*

$$\left\langle O^i \partial_\mu I^\mu{}_j \right\rangle_{\mathcal{C}} = -\mathrm{i}\hbar \left\langle O^i{}_{,j} \right\rangle_{\mathcal{C}} \tag{110}$$

where $O^i{}_{,j}$ is, again, the functional derivative of the observable with respect to the field. These identities encode the conservation of Noether currents at the quantum level (by taking $O = 1$) and constrain correlation functions. In non-Abelian gauge theories, they generalise to Slavnov–Taylor identities, which enforce gauge invariance after gauge fixing and incorporate ghost fields. Just as in the case of the Schwinger-Dyson equations, the integration should be taken on a space without boundaries so that the path integrals before and after change of variable can be identified.

Unfortunately, the path integral measure may not remain invariant under certain classical symmetries, producing a non-trivial Jacobian J. This produces an additional contribution and leads to the *anomalous Ward-Takahashi identities*

$$\left\langle O^i \partial_\mu I^\mu{}_j \right\rangle_{\mathcal{C}} = -\mathrm{i}\hbar \left\langle O^i{}_{,j} + O^i J_{,j} \right\rangle_{\mathcal{C}} \tag{111}$$

This new, last term is called an *anomaly* and it implies that classical symmetries are not necessarily preserved upon quantisation [89]. The path integral formalism provides a natural framework to compute these effects via the Jacobian of the change of variables.

In the context of worldline theories, a more technical but notable application is the *Duru–Kleinert transformation*, which turns seemingly complicated path integrals into exactly solvable ones. The trick consists in operating a functional change of variable that simultaneously reparametrises time (so $\mathcal{M}$) and the coordinates (so $\mathcal{N}$), starting with a path integral written in physical time t. A new parameter s is introduced, defined through a rescaling $\mathrm{d}t = g(q)\,\mathrm{d}s$ where $g(q)$ is a suitably chosen function of the configuration variables. Simultaneously, we perform a non-linear change of coordinates $f : q \mapsto Q$ with a new coordinate Q such that the action becomes quadratic in (Q, s). The canonical example is the hydrogen atom where $q = Q^2$ combined with $\mathrm{d}t = |Q|^2\,\mathrm{d}s$ maps the Coulomb potential onto a harmonic oscillator in Q. A singular problem in one set of variables becomes a free and exactly solvable theory in another [91].

However, the power of functional integration is that we can consider way more spectacular changes of variables. There is no reason why the diffeomorphism f should not map a history $\phi : \mathcal{M} \to \mathcal{N}$ to another history with completely different parameter and target spaces $\phi' : \mathcal{M}' \to \mathcal{N}'$. Of course, Jacobians might become arbitrarily difficult to compute, but at least symbolically, such changes of variables can shed light on the meaning of a quantum theory. It is very important to bear in mind that the configuration histories are *bound variables* of the quantum theory, and therefore may be completely changed without altering the physics. This is the path integral equivalent of changing the Hilbert space basis in Dirac formalism. Of course, the action changes accordingly, becoming f_*S (plus possible terms coming from the Jacobian) instead of S. Dramatic changes in the integration variables may thus lead to a different classical limit. This is a feature rather than a bug of the formalism: it is completely understandable that the classical limit should depend on a choice of classical variables. There are two examples that can help us imagine how this idea could be pushed.

Coherent state path integrals are a common example, usually introduced though a time-slicing procedure in which resolutions of the identity in the coherent state basis are used instead of the position basis. The resulting path integral is an integral over a space of pairs of conjugate normal variables, $\{\alpha(t), \alpha^*(t)\}$, with one such pair for each bosonic mode instead of one variable for each particle coordinate [92].

An even more dramatic example is the so-called *worldline formalism* of quantum field theories. In this formalism, the field variables of QFT (which live on $\mathcal{M}$ interpreted as d-dimensional spacetime) are substituted in the path integral for worldlines, that is, histories living on $\mathcal{M} \cong \mathbb{R}$ representing only time. Just as in non-relativistic quantum mechanics, the summation is taken over particle trajectories, but of course the action is radically different [93].

3.6 Effective field theory approach

Solving a physics problem often begins by trying to isolate the system under study from its larger environment, which implies taking into account complicated couplings with the rest of the universe, or effects arising from a more fundamental description of the system which we either do not know or do not want to describe. We would rather describe the system with an **effective theory** that reduces the environment to a fixed bath with which the system has some effective interaction. Basically, if ϕ_{sys} are the system's classical degrees of freedom, ϕ_{env} are the environment's degrees of freedom, $S_{\text{sys}}[\phi_{\text{sys}}]$ is the system's action, $S_{\text{env}}[\phi_{\text{env}}]$ is the environment's action, and $S_{\text{int}}[\phi_{\text{sys}}, \phi_{\text{env}}]$ encodes their arbitrary interactions, then we want to define the *effective action* $S_{\text{eff}}[\phi_{\text{sys}}]$ such that

$$\int \mathcal{D}\phi_{\text{sys}}\, \mathrm{e}^{\mathrm{i}S_{\text{eff}}[\phi_{\text{sys}}]/\hbar} = \int \mathcal{D}\phi_{\text{sys}}\, \mathcal{D}\phi_{\text{env}}\, \exp\left(\frac{\mathrm{i}}{\hbar}(S_{\text{sys}}[\phi_{\text{sys}}] + S_{\text{env}}[\phi_{\text{env}}] + S_{\text{int}}[\phi_{\text{sys}}, \phi_{\text{env}}])\right) \tag{112}$$

This fixes the effective action to be formally

$$S_{\text{eff}}[\phi_{\text{sys}}] = S_{\text{sys}}[\phi_{\text{sys}}] - \mathrm{i}\hbar \log \int \mathcal{D}\phi_{\text{env}}\, \exp\left(\frac{\mathrm{i}}{\hbar}(S_{\text{env}}[\phi_{\text{env}}] + S_{\text{int}}[\phi_{\text{sys}}, \phi_{\text{env}}])\right) \tag{113}$$

By computing the environment path integral, we have encoded its effect into a correction to the system' action. We have integrated out a sector of the theory, which really is an integration over the undesired degrees of freedom in the functional sense! Of course, the existence of such an exact effective action is not guaranteed. We know for a fact that path integrals are ill-defined, that even if they are well-defined they sometimes diverge, and that even if they converge they yield a complex result, so taking the log is ambiguous. It is however not a problem in practice: the whole point of effective theories is to avoid performing the exact computation of the integral (113). We rather

derive (or even postulate) an approximate form thereof, and then compute (exactly or through other approximation schemes) the remaining effective path integral on the l.h.s. of eq. (112), which is now only approximately equal to the exact path integral on the r.h.s.

A very straightforward example is the Born-Oppenheimer approximation in non-relativistic molecular physics, which consists in assuming all nuclei in a molecule to be fixed in the background while the electrons are treated as fully quantum. The usual justification of this approximation is that nuclei are much heavier than electrons and thus the characteristic time of electron dynamics is much shorter than the characteristic time of nucleus dynamics, decoupling the two sectors. This is however a classical argument, and we want to show how it can be made into a truly quantum argument. Let us note $\{r(t)\}$ the set of electron position histories[3], $\{R(t)\}$ the set of nucleus position histories[4]. The exact path integral is then

$$Z = \int \mathcal{D}r\, \mathcal{D}R\, \exp\left(\frac{\mathrm{i}}{\hbar}(S_e[r] + S_n[R] + S_{\mathrm{int}}[r,R])\right) \tag{114}$$

Since we are interested in finding the eigenstates of the molecule, we take that integration to the long-time limit so as to obtain the ground state. Recall that "long time" is defined as a time of integration $t \gg t_e$ with $t_e \sim 0.1$ fs the classical characteristic time of electron dynamics (*cf.* subsection 3.1). Now, because the classical characteristic time of *nucleus* dynamics $t_n \sim 10$ fs is larger, we can choose $t_n \gg t \gg t_e$. In this regime, by definition of t_n, nuclei have no time to move so any path deviating from the classical constant trajectory $R_c(t)$ will generate wild fluctuations of the action $\delta S \gg \hbar$, thus, according to the classical approximation of eq. (98):

$$\int \mathcal{D}R\, \exp\left(\frac{\mathrm{i}}{\hbar}(S_n[R] + S_{\mathrm{int}}[r,R])\right) \approx \exp\left(\frac{\mathrm{i}}{\hbar}(S_n[R_c] + S_{\mathrm{int}}[r,R_c])\right) \tag{115}$$

We immediately identify the effective action $S_{\mathrm{eff}}[r] = S_e[r] + S_n[R_c] + S_{\mathrm{int}}[r,R_c]$. Note that the approximation relies entirely on a choice of time scale, or equivalently, on a choice of energy scale. In systems where at least some electrons are slow or at least some nuclei are light, the time scales become similar, and further orders of semi-classical approximations may be needed to correct the effective action.

A slightly more involved example is provided by spontaneous symmetry breaking, a phenomenon usually introduced *classically* by pointing out that the classical minima of the action do not necessarily display the full symmetry of the action (imagine a Mexican hat potential as in fig. 4, which possesses rotational symmetry around its central axis, but whose minima lie in a circle: all minima are equivalent and there should be no general reason to fall in one rather than another). Moreover, this symmetry breaking is said to occur below a certain energy scale, as if the potential was changing its shape above this energy scale so that the minima become symmetric (the Mexican becoming *e.g.* a paraboloid as in fig. 4). This is only properly understandable in quantum mechanics, and the path integral expresses it quite simply.

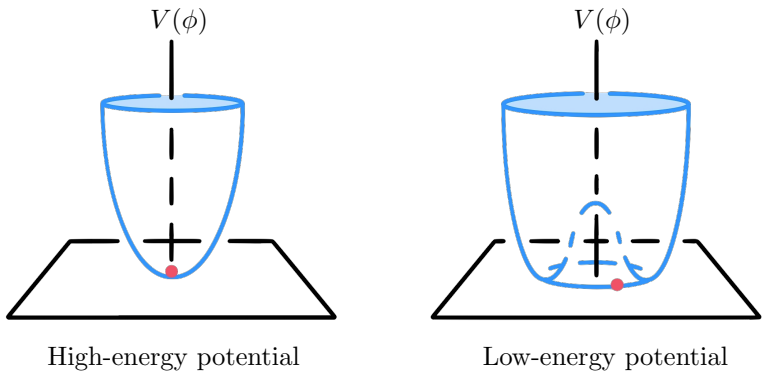

Figure 4: Mexican hat potential in the high- and low-energy limits. At high energy, the minimum (red dot) is symmetric under rotation around the potential axis, while at low energy, choosing one minimum on the circle of degenerate minima breaks the rotational symmetry of the potential.

[3]Since electrons are indistinguishable, $\{r\}$ is a complicated, non-linear, topological space where the permutation group acts trivially, *i.e.* $(\mathbf{r}_1,\ldots,\mathbf{r}_i,\ldots,\mathbf{r}_j,\ldots,\mathbf{r}_n) \sim (\mathbf{r}_1,\ldots,\mathbf{r}_j,\ldots,\mathbf{r}_i,\ldots,\mathbf{r}_n)$, but this is not important in the discussion.

[4]The same remark applies if the molecule contains the same nucleus several times.

First, choose one of the degenerate classical histories ϕ_c and operate a change of integration variable $\phi \mapsto \eta = \phi - \phi_c$. At low energies, the discussion becomes similar to the case of the Born-Oppenheimer approximation: only the near-classical histories contribute and the system is described at first order in $\hbar$ by the effective action $S[\phi_c] - \mathrm{i}\hbar \log \int \mathcal{D}\eta \exp\big(\mathrm{i}\eta^i S_{,ij}\eta^j/2\big)$, which comes from the semi-classical expansion (98). In this sense, the choice of a particular classical history ϕ_c *spontaneously breaks* the symmetry of the action (there is no reason why the symmetries of the full action should remain symmetries of the effective action). Above the symmetry-breaking scale, however, the semi-classical expansion becomes asymptotic and the effective action recovers the full symmetries. For instance, the effective action of the Higgs field at low energies contains masses for the electroweak bosons that are forbidden in the full action. Note that spontaneous symmetry breaking need not occur at the classical level (zeroth order in $\hbar$): it may happen at some higher, semi-classical order, but still the symmetry will be restored in the asymptotic limit. This case is dubbed *dynamical symmetry breaking*, and per this discussion should not be seen as fundamentally different from "classical" spontaneous symmetry breaking.

The systematic generalisation of these scale arguments is the subject of Effective Field Theories (EFTs), whose aim is to describe a physical system at an energy scale E while systematically accounting for the heavier degrees of freedom even without full knowledge of the high-energy (UV) theory. For a more complete but non exhaustive overview of EFTs, see for example refs. [94, 95, 96].

In the full quantum theory, processes occurring at an energy scale E are insensitive to the precise dynamics of the heavy sector, defined as fields with masses $Mc^2 \gg E$, and the path integral provides techniques to achieve this decoupling [97]. By integrating out the heavy fields and focussing only on the relevant degrees of freedom at a given scale, the effects of the heavy fields are encoded into higher-dimensional operators in the Lagrangian and renormalized couplings.

Consider for example some quantum field theory with light fields ℓ with mass $mc^2 \ll E$ and heavy fields H with mass $Mc^2 \gg E$, described as before by the action

$$S[\ell, H] = S_{\text{light}}[\ell] + S_{\text{heavy}}[H] + S_{\text{int}}[\ell, H] \tag{116}$$

In this example, low-energy physics is defined as the scale of the fields ℓ with typical momenta $p^2 \ll M^2c^2$, and the low-energy effective action $S_{\text{eff}}[\ell]$ is defined by integrating the heavy sector of the theory. So far, this procedure is completely exact. Let us note that the resulting effective action is generally non-local because the integral over H will generate interactions mediated by heavy-field propagators that will couple ℓ at different points of spacetime. To see it, suppose that the action of eq. (116) is quadratic in the heavy fields

$$S[\ell, H] = S_{\text{light}}[\ell] + \int \mathrm{d}^4x \left[\frac{1}{2}H\left(-\Box - \frac{(Mc)^2}{\hbar^2}\right)H + HJ(\ell)\right] \tag{117}$$

where $\Box$ is the d'Alembert operator and $J(\ell)$ some functional source built from the light fields. Since we assumed the heavy field appears quadratically, performing the integration yields

$$S_{\text{eff}}[\ell] = S_{\text{light}}[\ell] + \frac{1}{2}\int \mathrm{d}^4x\, \mathrm{d}^4y\, J(x)\Delta_F(x-y, M)J(y) \tag{118}$$

where we have denoted the heavy propagator as Δ_F. As a result, the effective action contains an interaction term that is manifestly non-local (it connects two different points of spacetime). However, in the low-energy limit, the propagation of heavy particles is exponentially suppressed by their mass, so we can expand the heavy-field propagator in powers of $p^2/(Mc)^2$

$$\frac{1}{p^2 - (Mc)^2} = -\frac{1}{(Mc)^2}\left(1 + \frac{p^2}{(Mc)^2} + \dots\right) \tag{119}$$

and Fourier-transform to position space:

$$-\frac{1}{(Mc)^2}\left(\delta^{(4)}(x-y) - \frac{\hbar^2\Box}{(Mc)^2}\delta^{(4)}(x-y) + \dots\right) \tag{120}$$

The expression above is a sum of local operators that we generally organize as a tower of operators suppressed by the heavy scale M

$$S_{\text{eff}}[\ell] = \int \mathrm{d}^4x \left[\mathcal{L}_4(\ell) + \sum_{d\geq 5}\left(\frac{\hbar}{Mc}\right)^{d-4} c_d O_d(\ell)\right] \tag{121}$$

where $\mathcal{L}_4(\ell)$ is the Lagrangian density containing operators of mass dimension smaller than 4, $O_d(\ell)$ are local operators of mass dimension d built from light fields and c_d are Wilson coefficients of the expansion. The Wilson coefficients encode what we do not know about the heavy field dynamics. If we know the exact UV dynamics of the system, they are determined by computing a set of observables in the full theory and in the EFT framework and making them match at a specific energy scale, but if not, they are treated as free parameters to be fit on experiment. Crucially, there are generally several different operators for a given dimension d. For example, if there are any heavy particles beyond the description of the Standard Model of particle physics, the path integral allows to write Standard Model EFTs (SMEFTs) and encode the influence of heavy particles into Wilson coefficients. This way, we translate the experimental absence of new particles into precise UV constraints on the theory [98]. A representation is given in fig. 5.

The textbook example of effective theories may be the V-A (Vector - Axial vector) theory of weak interaction, proposed by Fermi in 1933/1934 [99]. The mediators of the weak interaction between two fermions are massive W, Z bosons, and at energies much lower than their masses, their propagation is suppressed. In this limit, the effective interaction can be captured by a contact term between fermions, without explicitly including the heavy mediator, and their effects are encoded in a series of local operators suppressed by the mass scale.

To cite another example, in the Ginzburg-Landau theory of superconductivity [100], electrons are treated collectively with the use of an order parameter rather than keeping track of all of them. This way, the theory directly accounts for the macroscopic superconducting state.

The discussion we had on lattice field theory in subsection 3.2 is also an example of effective theory, precisely because the lattice spacing is a high-energy cutoff. The lattice action can be seen as an effective description of the system, holding as long as momenta are much smaller than the inverse spacing. The low-energy physics can be computed up to corrections in powers of the spacing, and the choice of discretisation procedure is the direct parallel of fixing a set of Wilson coefficients.

In each of these constructions, the path integral reveals that when degrees of freedom are separated by a large hierarchy of scales, the functional measure allows to reorganize the theory involving only the relevant low-energy variables. Integrating out a sector of the theory is literally an integration in the functional sense and the influence of heavy modes can be parametrised by a systematic expansion into local operators. Of course, we have seen in the previous subsection that the choice of integration variable in the path integral is completely arbitrary, and suitable changes of variables can sometimes be made to transform arbitrary variables ϕ_1, ϕ_2 into "separable" ones ℓ, H that belong to sectors separately integrable sectors. A general effective theory is thus the combination of

- a well-chosen change of variable that dispatches the quantum information of the theory into physically (quasi-)isolated sectors;
- an approximate integration of the "uninteresting" degrees of freedom giving rise to a new, smaller theory only describing the sector of interest.

These two steps can in principle be iterated and various methods can be used at each iteration to find the effective action. As one progressively integrates out modes at successively lower and lower energy scales, the flow of successive effective actions can be encoded in differential equations that describe how each coupling evolves with scale. This is precisely the study of the renormalisation group (for a detailed introduction, see [101]).

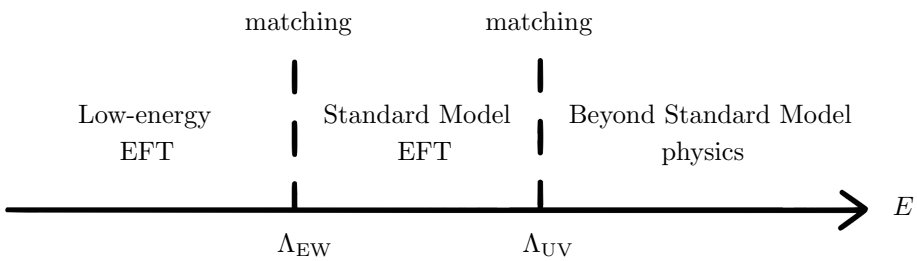

Figure 5: Example in the Standard Model: above some high-energy Λ_{UV} scale lies some unknown physics we can choose to systematically parametrize using EFT techniques. The same can be done for low-energy physics (typically below the electroweak scale $\Lambda_{\mathrm{EW}} \sim 246$ GeV) to encapsulate the effects of heavy SM degrees of freedom in low-energy observables. Wilson coefficients can be computed in each regimes, and matching conditions have to be applied to ensure consistency and continuity between scales.

4 Some proofs in the symbolic calculus framework

In this section, we would like to present some illustrative examples of symbolic calculus using path integrals, including the Hellmann-Feynman theorem, aspects of the Quantum Chromodynamics vacuum, and T-duality in string theory. The examples we choose are broad and heterogeneous by choice in order to highlight that path integrals can provide insights over a wide range of contexts.

4.1 The Hellmann-Feynman theorem

In many physical cases, we are interested in how observables respond to some external parameter λ. If we assume that the classical action and the observable functionals explicitly depend on λ, then path integrals provide analytical formulae for the derivative of expectation values with respect to this external parameter. Such derivatives can be used to obtain potential energy surfaces and other observable quantities.

Let us begin by differentiating the partition function

$$\frac{\mathrm{d}Z}{\mathrm{d}\lambda} = \frac{\mathrm{i}}{\hbar}\int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar}\frac{\mathrm{d}S}{\mathrm{d}\lambda} = Z\frac{\mathrm{i}}{\hbar}\left\langle \frac{\mathrm{d}S}{\mathrm{d}\lambda}\right\rangle \tag{122}$$

Now, we differentiate the expectation value of some observable $\langle O\rangle$, taking into account that Z, S and O all depend on λ:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle O\rangle &= \frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\frac{1}{Z}\int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar}O\right] \\
&= \frac{1}{Z^2}\int \mathcal{D}\phi\left(Z\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\mathrm{e}^{\mathrm{i}S/\hbar}O\right] - \frac{\mathrm{d}Z}{\mathrm{d}\lambda}\mathrm{e}^{\mathrm{i}S/\hbar}O\right) \\
&= \frac{1}{Z^2}\int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar}\left[Z\frac{\mathrm{i}}{\hbar}\frac{\mathrm{d}S}{\mathrm{d}\lambda}O + Z\frac{\mathrm{d}O}{\mathrm{d}\lambda} - \frac{\mathrm{d}Z}{\mathrm{d}\lambda}O\right] \\
&= \frac{1}{Z}\int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar}\frac{\mathrm{d}O}{\mathrm{d}\lambda} \quad + \quad \frac{\mathrm{i}}{\hbar Z}\int \mathcal{D}\phi\, \mathrm{e}^{\mathrm{i}S/\hbar}\left[\frac{\mathrm{d}S}{\mathrm{d}\lambda}O - \left\langle\frac{\mathrm{d}S}{\mathrm{d}\lambda}\right\rangle O\right] \\
&= \left\langle\frac{\mathrm{d}O}{\mathrm{d}\lambda}\right\rangle + \frac{\mathrm{i}}{\hbar}\left(\left\langle\frac{\mathrm{d}S}{\mathrm{d}\lambda}O\right\rangle - \left\langle\frac{\mathrm{d}S}{\mathrm{d}\lambda}\right\rangle\langle O\rangle\right)
\end{aligned} \tag{123}$$

where we recognise in the last term the covariance of $\mathrm{d}S/\mathrm{d}\lambda$ and $\langle O\rangle$.

In particular, if we take O to be the energy E and take the path integrals to the long-time limit, we obtain that all expectation values in eq. (123) are stationary state expectation values. Yet E is exactly known in any stationary state (in linear algebra, a stationary state is an energy eigenstate; in probabilistic terms, the energy is a constant random variable), so it has zero correlation to any other observable, thus for any stationary state:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle E\rangle = \left\langle\frac{\mathrm{d}E}{\mathrm{d}\lambda}\right\rangle \tag{124}$$

a result known as the Hellmann-Feynman theorem [102]. Eq. (123) then reveals that the "finite-temperature correction" to the Hellmann-Feynman theorem is the covariance of E and $\mathrm{d}S/\mathrm{d}\lambda$.

By contrast with the traditional proof, this derivation points out the variational nature of the Hellmann-Feynman theorem, which is especially important when considering degenerate stationary states: the long-time principle *continuously* selects a stationary state for each value of λ and ignores degeneracies, which allows the theorem to remain valid [103]. We show in the next subsection how degeneracies can be recovered in some path integrals.

4.2 The QCD vacuum

We would now like to demonstrate that path integrals can be applied to more intricate vacuum structures. We consider the construction of Quantum Chromodynamics (QCD), the theory that describes quarks, gluons and the strong interaction. Without loss of generality, the coupling constant of the interaction is set to one. While traditional proofs of the structure of the QCD vacuum rely on the introduction of a Bloch wave to interpret and obtain the precise vacuum structure [104], we propose here a pure path integral-motivated argument.

The dynamical variable of the action is the non-Abelian gauge field $A_\mu(x) = A^a_\mu(x)t^a$, where μ is the spacetime index, and t^a are the generators of $\mathfrak{su}(3)$. We define the non-Abelian field strength as $F_{\mu\nu} = \partial_{[\mu}A_{\nu]} - \mathrm{i}[A_\mu, A_\nu]$, which is a generalisation of the electromagnetic case.

As discussed in subsection 3.1, the vacuum is extracted from the long-time limit, which tells us what we are computing but says nothing about its precise structure. To uncover it, we impose the natural requirement that the configuration dominating the vacuum path integral carries finite energy, or equivalently finite action $S < \infty$. This requirement implies that the field strength vanishes at infinity $F_{\mu\nu}(x) \to 0$, which forces the gauge field to become pure gauge $A_\mu \to \mathrm{i}U\partial_\mu U^{-1}$ with $U(x) \in \mathrm{SU}(3)$ in order for the electric and magnetic fields to vanish. The gauge field at each point of infinity is thus completely determined by an element of the gauge group SU(3).

This condition means that gauge fields are then defining maps $A_\mu : S^3_\infty \to \mathrm{SU}(3)$, where S^3_∞ is the sphere at spacetime infinity. Interestingly, this means that there exist topologically inequivalent finite-action gauge fields that cannot be continuously deformed into each other. Topologically inequivalent gauge fields are said to belong to different homotopy classes, and the set of homotopy classes is the homotopy group $\pi_3(\mathrm{SU}(3)) \cong \mathbb{Z}$ (for an introduction to homotopy theory and its rich physical implications, see [105, 106]).

$\pi_3(\mathrm{SU}(3)) \cong \mathbb{Z}$ means that homotopy classes are labelled by a topological charge $w \in \mathbb{Z}$ called the "winding number", which counts how many times the 3-sphere wraps around the SU(3) manifold. Because of this topology, the total space of histories decomposes into disconnected sectors labelled by the winding number, and for each configuration history A with winding $w[A]$, we may define a past winding m and a future winding n such that $w[A] = n - m$. The full path integral then decomposes as a sum over all possible past and future windings

$$\int \mathcal{D}A\, \mathrm{e}^{\mathrm{i}S[A]/\hbar} = \sum_{m,n} \int_{w[A]=n-m} \mathcal{D}A\, \mathrm{e}^{\mathrm{i}S[A]/\hbar} \tag{125}$$

Such (past and future) configurations carrying different windings are classically separated by finite-energy potential barriers. The presence of transition amplitudes between different winding configurations in the quantum theory (the terms with $w[A] \neq 0$ in eq. (125)) indicate the presence of tunnelling events [107].

For the sake of curiosity, let us replace the sum of eq. (125) by a weighted sum, with a real parameter θ comprised between 0 and 2π:

$$\begin{aligned} \sum_{m,n} \mathrm{e}^{-\mathrm{i}\theta(n-m)} \int_{w[A]=n-m} \mathcal{D}A\, \mathrm{e}^{\mathrm{i}S/\hbar} O[A] &= \int \mathrm{d}k\, \mathrm{e}^{-\mathrm{i}k\theta} \int \mathcal{D}A\, \mathrm{e}^{\mathrm{i}S/\hbar} \delta(k - w[A]) O[A] \\ &= \int \mathcal{D}A\, \mathrm{e}^{\mathrm{i}S/\hbar - \mathrm{i}\theta w} O[A] \end{aligned} \tag{126}$$

where we rewrote the sum as an integral over Dirac deltas in the first line.

We see that summing over topological classes with phase $\mathrm{e}^{-\mathrm{i}k\theta}$ is exactly equivalent to a single integral with modified action. Crucially, the term θw does not depend on time! Recall from analytical mechanics that the energy is given by a derivative of the action [108]

$$\langle E \rangle = -\left\langle \frac{\partial S}{\partial t} \right\rangle \tag{127}$$

and we conclude that the additional θ-term we obtained contributes zero net energy. The theory thus admits a degenerate set of vacua, each specified by a value of θ and called a θ-vacuum. Note that we did not need the precise form of the action or any detail on the dynamics of quarks and gluons to deduce the form of the ground state. Moreover, including this θ-term in the action is completely equivalent to working in the θ-vacuum in the Dirac formalism [109]. Once interpreted in the QCD framework, this additional contribution violates the CP (charge-parity) symmetry, this is the so-called *strong CP problem*, and leads to the axion solution [110].

4.3 T-duality in string theory

We now turn to a framework that is a bit different from the ones we considered before, in the sense that the fundamental objects are not point-like particles anymore. In string theories, the parameter space $\mathcal{M}$ is taken to be a 2-dimensional worldsheet, and the target space $\mathcal{N}$ is often identified with spacetime. Here, a history describes the propagation of a string, and its vibrational modes generate the observed particles and forces. In this context, *T-duality* is defined as an equivalence relation between two different classical string theories, one that is compactified on a circle of radius R and another one on a circle of radius α/R, with α the square of the string characteristic length scale. Compactification means that one spatial dimension is constrained with a periodic condition $x \sim x + 2\pi R$ so that it effectively forms a circle. For a point particle, such compactification causes

the spectrum to develop Kaluza-Klein modes with typical momentum n/R, $n \in \mathbb{N}$ [111, 112], but a string can, in addition to these modes, wind k times around the compact direction at an energy cost kR/α. T-duality corresponds to the formal exchange between n and k.

Usual understanding of how T-duality arises requires expanding all modes and checking if the full spectrum is invariant. Here, following the elegant proof of Tong [113], we want to underline how path integral makes this duality manifest.

Consider a free scalar field ϕ defined on a Euclidean worldsheet Σ parametrized by two parameters Σ^μ, $\mu = 1, 2$. The field takes values on a circle of radius R such that we identify $\phi \sim \phi + 2\pi$ under periodicity. The Euclidean path integral is then

$$Z_E = \oint \mathcal{D}\phi \, \mathrm{e}^{-S_E[\phi]/\hbar}, \qquad S_E[\phi] = \frac{\hbar c R^2}{4\pi\alpha} \int_\Sigma \mathrm{d}^2\Sigma \; \partial_\mu \phi \partial^\mu \phi \tag{128}$$

The claim of T-duality translates as $Z_E(R) = Z_E(\alpha/R)$ up to a proper integration on the fields.

To be able to perform the computation, we must first note that the action is symmetric under constant shift $\phi \mapsto \phi + \chi$. To deal with such a symmetry, we promote χ to an arbitrary field on the worldsheet and introduce a U(1) gauge field A_μ transforming as $A_\mu \to A_\mu - \partial_\mu \chi$. Ordinary derivatives are then replaced by covariant derivatives $D_\mu \phi = \partial_\mu \phi + A_\mu$. To ensure that this gauge theory is not larger than the previous one because of gauge symmetry, we apply what we have learned in subsection 3.5 and introduce a Lagrange multiplier λ such that the action writes

$$S'_E[\phi, A, \lambda] = \frac{\hbar c R^2}{4\pi\alpha} \int_\Sigma \mathrm{d}^2\Sigma \; D_\mu \phi D^\mu \phi + \frac{\mathrm{i}\hbar c}{2\pi} \int_\Sigma \mathrm{d}^2\Sigma \; \lambda \varepsilon^{\mu\nu} \partial_\mu A_\nu \tag{129}$$

Integrating over λ enforces $\varepsilon^{\mu\nu}\partial_\mu A_\nu = 0$ and correctly recovers the original theory of eq. (128). The partition function is then rewritten as

$$Z_E = \frac{1}{V} \oint \mathcal{D}\phi \, \mathcal{D}A \, \mathcal{D}\lambda \, \mathrm{e}^{-S'_E[\phi, A, \lambda]/\hbar} \tag{130}$$

where the volume V of the gauge group is a divergent prefactor that compensates the divergent overcounting of gauge-equivalent configurations. Rigorously dealing with the normalisation requires care in this example, we will not deal much with the precise considerations because the final result can be redefined up to proper rescaling without alterning the logic nor conclusion. In the integral, ϕ appears only through its covariant derivative, and we see that shifting ϕ by a constant and at the same time shifting A_μ by the same amount produces the same $D_\mu \phi$. It means that many different pairs (ϕ, A_μ) give the same observable, they are all equivalent, so we can prevent overcounting by choosing one specific pair. We can fix the gauge to $\phi = 0$, eliminating $\mathcal{D}\phi$, and absorbing the integral in ϕ inside the volume prefactor:

$$Z_E = \oint \mathcal{D}A \, \mathcal{D}\lambda \, \exp\left(-\frac{cR^2}{4\pi\alpha} \int_\Sigma \mathrm{d}^2\Sigma \; A_\mu A^\mu + \frac{\mathrm{i}c}{2\pi} \int_\Sigma \mathrm{d}^2\Sigma \; \varepsilon^{\mu\nu} (\partial_\mu \lambda) A_\nu \right) \tag{131}$$

The last term is obtained after integration by part assuming a worldsheet with no boundary. The action is now quadratic in the gauge field, allowing use to complete the square by explicitly rewriting:

$$Z_E = \oint \mathcal{D}A \, \mathcal{D}\lambda \, \exp\left(-\frac{cR^2}{4\pi\alpha} \int_\Sigma \mathrm{d}^2\Sigma \left(A_\mu + \frac{\mathrm{i}\alpha}{R^2} \varepsilon_{\mu\nu} \partial^\nu \lambda \right)^2 - \frac{c\alpha}{4\pi R^2} \int_\Sigma \mathrm{d}^2\Sigma \partial_\mu \lambda \partial^\mu \lambda \right) \tag{132}$$

The first integral is Gaussian, we can perform a functional change of variable in the form of a generic linear shift here, by defining

$$\tilde{A}_\mu = A_\mu + \frac{\mathrm{i}\alpha}{R^2} \varepsilon_{\mu\nu} \partial^\nu \lambda \tag{133}$$

Because this is a shift, $\mathcal{D}\tilde{A} = \mathcal{D}A$, the result of the Gaussian integration over $\tilde{A}_\mu$ is a prefactor that is field-independent and it can be absorbed into normalisation so that the entire computation reduces in the very end to an integration over a scalar field

$$Z_E = \oint \mathcal{D}\lambda \, \exp\left(-\frac{c\alpha}{4\pi R^2} \int_\Sigma \mathrm{d}^2\Sigma \, \partial_\mu \lambda \partial^\mu \lambda \right) = \oint \mathcal{D}\lambda \, \exp\left(-\frac{c\tilde{R}^2}{4\pi\alpha} \int_\Sigma \mathrm{d}^2\Sigma \, \partial_\mu \lambda \partial^\mu \lambda \right) \tag{134}$$

with $\tilde{R} = \alpha/R$. Comparing with the initial path integral, we see that we have performed two replacements $\phi \mapsto \lambda$, $R \mapsto \tilde{R}$ and that it keeps the exact same form. Path integrals offer a very

convenient mean to derive this duality: the explicit dependence on the classical action allowed us to gauge the classical action (but in a constrained way so that the quantum theory remained unchanged) and then ungauge it by integrating over the original string. This clearly illustrates the unphysical character of the integration variables than may be swapped without altering the overall physics.

In some sense, this duality contrasts with the anomaly: instead of having a classical symmetry that fails to extend to the quantum realm, we see that two quite different classical theories (respectively expressed by eqs. (128) and (134)), related by a drastic transformation $R \mapsto \alpha/R$, in fact yield the very same quantum theory. From a fundamental point of view, this clearly shows that the concept of classical limit is somewhat non-universal and completely depends on a choice of classical action inside the path integral. From a practical point of view, this opens the way to complete redefinitions of quantum theories, which could be used to perform calculations initially thought impossible (sending the strong coupling limit to the perturbative limit in terms of a dual coupling constant, for instance). Finally, and looking forward, this could suggest that many apparently different classical theories may be mere avatars (distinct classical limits) of a unique underlying quantum theory.

Conclusion

We hope that this review convinced the reader that path integrals are useful, robust and insightful tools when considering either the quantisation of a classical theory (be they Newtonian or relativistic, worldline or field), or the modelling of a given quantum system. In many cases, path integrals provide a clearer understanding of somewhat cumbersome quantum concepts, and they sometimes have guided physicists towards the conception of new tools and theories.

Path integrals remain a weird mixture of physical intuition and many different mathematical frameworks, which makes them both desperately elegant and very difficult to grasp in detail. Further developments in the mathematics of path integrals are expected and would be warmly welcomed by the community, in order to both guarantee the rigorous validity of the path integral calculus, draw its limits, and possibly perform the complete quantisation of certain theories that remain to this day out of our comprehension. The most famous example thereof is the quantisation of Yang-Mills theories (for the description of strong nuclear interactions for instance), whose non-perturbative formulation is still unknown and constitute one of the major mysteries of modern mathematical physics [55].

Finally, we firmly believe that being able to symbolically operate on path integrals means being able to express many different quantum-mechanical methods in a unified way, which might allow for systematic comparisons of these methods and fine control over the possible approximations they imply. We namely wrote the sections on perturbative expansions and changes of variables with this idea in mind: to show they were special cases (or special steps) of the general framework of effective theories. In this sense, any problem in quantum chemistry, strongly-correlated condensed matter, nuclear physics or quantum chromodynamics, for instance, should be seen as a succession of suitable changes of variables that condense the information in relevant classical variables, and of suitable approximations that allow for the computation of a relevant effective action, until the problem becomes simple enough that it can be completely solved. In the best of cases, this programme could lead to systematic improvements of current methods. At the very least, it would allow for the comparison and maybe classification of these current methods, and would simplify their translation from one field of research to another.

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Author contributions

The authors contributed equally to the research and share first authorship.

Data availability

No new data were created or analysed in this study.

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